Journal of Experimental Psychology: Learning, Memory, and Cognition

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Daniel Fitousi

Online First Publication, May 18, 2023. https://dx.doi.org/10.1037/xlm0001250

CITATION
Fitousi, D. (2023, May 18). Dimensional Interaction Is Driven by Feature Integration: A Binding Theory of Garner Interference. *Journal of Experimental Psychology: Learning, Memory, and Cognition* Advance online publication. https://dx.doi.org/10.1037/xlm0001250

Journal of Experimental Psychology:
Learning, Memory, and Cognition

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ISSN: 0278-7393
https://doi.org/10.1037/xlm0001250

Dimensional Interaction Is Driven by Feature Integration: A Binding Theory of Garner Interference

Daniel Fitousi
Department of Psychology, Ariel University

For nearly half a century now, Garner interference has been serving as the gold standard measure of dimensional interaction and selective attention. But the mechanisms that generate Garner interference are still not well understood. The current study proposes a novel theory that ascribes the interference (and dimensional interaction in general) to episodic feature integration processes at the micro (trial-to-trial) level. The novel account builds on earlier well-established notions of "feature integration" and "object files," and is augmented by formal derivations. The sequential binding account predicts that the magnitude of Garner interference is related to the strength of feature integration along consecutive trials. Three experiments were set to test this novel binding theory. Experiment 1 and Experiment 2 tested performance with integral dimensions (chroma and value, and width and height of rectangles); whereas Experiment 3 examined performance with a pair of separable dimensions (circle's size and angle of a diameter). In addition, the time lag ensuing between consecutive trials was manipulated. The results strongly supported the predictions of the sequential binding account: (a) with integral dimensions, substantial amounts of Garner interference were correlated with large partial repetition costs (e.g., consensual markers of feature integration), but this pattern was not observed with separable dimensions, and (b) the magnitude of both Garner interference and partial repetition costs diminished as a function of the ensuing time lag between consecutive trials, pointing to a common time-dependent memory mechanism. These results adduce strong support in the predictions of the feature binding theory of Garner interference, giving currency to the idea that dimensional interaction is driven by feature integration.

Keywords: Garner speeded classification task, sequential effects, feature binding

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The world that surrounds us is inhabited by multidimensional objects (e.g., cars, trees, and faces). These are composed of various dimensions (e.g., color, shape, and size). People can notice these attributes quite easily, extracting the *color* of an approaching car or its *size*. These effortless achievements often support the belief that dimensions are independent entities. However, those naive beliefs are shattered when people perform the simplest of selection tasks, such as that of naming the ink color of a color word (e.g., BLUE in green) while ignoring its meaning (Stroop, 1935); or that of judging the width of a rectangle while ignoring its height (Weintraub, 1971). In these cases, the constituting dimensions exhibit some level of cross-talk. How strong is that cross-talk? This question of dimensional interaction is intertwined with another fundamental caveat in psychology—the selectivity of attention (Treisman & Gelade, 1980). Can observers focus their attention on one dimension of an object and ignore its other irrelevant dimension? A pillar of modern cognitive science has been Wendell R. Garner's (Garner, 1974, 1976; Garner & Felfoldy, 1970) distinction between *integral* and *separable* dimensions. Objects that are composed of integral dimensions, such as patches of color that vary in hue and saturation (Garner & Felfoldy, 1970) are processed in their totality as wholes. In such objects, selective attention to the relevant dimension (hue) is compromised due to irrelevant variation on the other dimension (saturation). Such dimensions are considered to be dependent in processing (Hyman & Well, 1967; Lockhead, 1972). In contrast, objects that are composed of separable dimensions, such as circles varying in size and a diameter varying in angle drawn within them (Shepard, 1964) allow for good selective attention to the criterial dimension (e.g., circle's size). Such dimensions are considered to be independent in perception (Ashby & Townsend, 1986). The current study postulates that the level of dimensional interaction is strongly related to differential amounts of binding between their constituent features. Thus, the main claim of the present effort is that dimensional interaction is driven by feature integration.

A popular methodology for assessing the level of dimensional interaction and selectivity of attention with pairs of integral and separable dimensions is *Garner's speeded classification paradigm* (Garner, 1974, 1976; Garner & Felfoldy, 1970), which entails intrusions by task-irrelevant dimension on performance with the task-relevant dimension. In the Garner task, the participant classifies the values of a relevant dimension, while ignoring those of an irrelevant dimension. *Garner interference* is computed as difference in performance between two experimental blocks: in *filtering*, the irrelevant dimension varies orthogonally from trial-to-trial, whereas in *baseline*, the irrelevant dimension is held fixed at one value. The larger the

Daniel Fitousi thanks Dr. Bernhard Hommel and two anonymous reviewer for their insightful comments on an earlier version of this paper. Fitousi also thanks Omer Azizi for running Experiments 2 and 3. This research was supported by the Israel Science Foundation (Grant 1498/21). All data can be downloaded from here (Fitousi, 2023).

Correspondence concerning this article should be addressed to Daniel Fitousi, Department of Psychology, Ariel University, Kiryat Hamda, 3, Ariel, 40700, Israel. Email: danielfi@ariel.ac.il

magnitude of Garner interference (GI), the greater is the failure of selectivity to the criterial dimension. Since its inception, Garner's paradigm and its attendant conceptions have been applied successfully to a wide range of research topics spanning from categorization (Ashby & Maddox, 1994; Ashby & Perrin, 1988; Nosofsky, 1986, 1987), to shape perception (Pomerantz, 1983; Pomerantz et al., 2003), to face recognition (Amishav & Kimchi, 2010; Fitousi, 2015), to numerical cognition (Algom et al., 1996), to perceptual development (Shepp & Swartz, 1976; Smith & Kemler, 1978), to the perception of sound (Melara & Marks, 1990a, 1990b), to name only a few examples of research topics that benefited from Garner's paradigm and its attendant concepts (for a comprehensive review see, Algom & Fitousi, 2016). However, alongside its immense utility in unearthing cognitive processing, one question has pervasively stood up in Garner research. This pressing question has been succinctly put by Garner (1974) himself: "What causes interference?" (p. 140). Two major explanations have been proposed to account for the emergence of Garner interference—*irrelevant variation* (Felfoldy, 1974; Garner, 1974) and—*stimulus uncertainty* (Ashby & Maddox, 1994). Several ingenious attempts have been made (Ashby & Maddox, 1994; Burns, 2016; Dyson & Quinlan, 2010; Liu et al., 2022; Niv et al., 2022; Nosofsky & Palmeri, 1997) to decide between the two alternatives. However, these approaches are not really explanations of the effect but, rather, try to pinpoint the actual problem behind the effect. True, pinpointing the actual problem is an important step in finding out the underlying mechanism, but it does not already present such a mechanism. Thus, the root cause of GI, and hence the nature of selectivity with separable and integral dimensions, still remain to be discovered.

The current study outlines a new theory of GI, one that relies on basic mechanisms of feature binding in episodic memory (Hommel's, 1998; Hommel, 2004; Kahneman et al., 1992). According to this novel account, dimensional interaction is driven by feature integration. Here, the integral-separable distinction, and their associated selective attention patterns are attributed to differences in the strength of integration among the constituent features of the pertinent dimensions. When features are more strongly bound, they are also more difficult to tease apart. Feature binding occurs for both integral and separable dimensions, but the strength of binding is shown to be much larger with integral dimensions. This can also explain the failure of selective attention to the relevant dimension, which, in return, gives rise to GI. Binding in the Garner paradigm is demonstrated by the presence of unique sequential effects (partial repetition costs) across consecutive trials that reflect a binding mechanism (Hommel's, 1998). The new theory places dimensional interaction within the rich context of feature binding theories (Hommel's, 1998; Kahneman et al., 1992). The goals of the current study are therefore twofold: (a) to provide the theoretical basis for the relations between GI and sequential feature binding, and (b) present empirical support for this new theory. The current study sheds new light on the nature of dimensional interactions and their relations to feature integration.

Selective Attention and Dimensional Interaction

The objects of perception are multidimensional. We are often required to attend selectively to one dimension while ignoring the other dimension. A prime laboratory example of this phenomenon is the Stroop effect (SE) (Stroop, 1935). When naming the ink color in which color words are written, people fail to ignore the meaning of the carrier words. To probe the impact of the task-irrelevant words, the SE is computed as the difference in color naming performance between incongruent (e.g., RED in green) and congruent (e.g., RED in red) stimuli.

$$\mathrm{SE} = E(\mathrm{RT}|\mathrm{incongruent}) - E(\mathrm{RT}|\mathrm{congruent}), \quad (1)$$

where $E(\mathrm{RT}|\mathrm{incongruent})$ and $E(\mathrm{RT}|\mathrm{congruent})$ are the expected values (means) of response times to name the ink color with incongruent and congruent stimuli, respectively. The Stroop effect entails failure of selective attention to the relevant dimension (ink color) due to conflict with the irrelevant dimension (color name). Attention is compromised due to dimensional cross-talk or interaction between the color dimension and the word dimension. However, we note that Equation 1 cannot be deployed with every pair of dimensions. Apparently, not all multidimensional stimuli are also Stroop stimuli (Algom & Fitousi, 2016; Algom et al., 2022; Melara & Algom, 2003). That is, the quality of congruency does not apply in most stimuli. Think for example about a red triangle or a green circle. There is no congruency or incongruency relations between the dimensions of color and shape. Thus, the Stroop paradigm is restricted only to pairs of dimensions to which this quality applies. Unfortunately, the great majority of dimensions in our world do not own this quality. A major advantage of the Garner paradigm is that it can be used to assess selective attention with any pair of dimensions, Stroop or non-Stroop.

Garner's Speeded Classification Task

Consider the application of the paradigm to the dimensions of color and shape. The observer is presented with a shape (either a circle or a square) and has to decide, while timed, on the shape, and ignore its color (which can be either red or green). The design consists of two primary experimental blocks. In *baseline*, the task-relevant dimension (shape) varies from-to-trial, while the task-irrelevant (color) is held constant at one value (e.g., red). In *filtering*, both task-relevant and task-irrelevant dimensions vary randomly from trial-to-trial. The difference in mean RT between performance in these two blocks defines the measure known as *Garner interference* (GI) (Pomerantz, 1983):

$$\mathrm{GI} = E(\mathrm{RT}|\mathrm{filtering}) - E(\mathrm{RT}|\mathrm{baseline}), \quad (2)$$

where $E(\mathrm{RT}|\mathrm{filtering})$ and $E(\mathrm{RT}|\mathrm{baseline})$ are expected values of RTs for classifying the relevant dimension in filtering and baseline, respectively. An analogue formula exists for accuracy. The presence of GI implies that irrelevant variation on color took a toll on the performance with the shape, and that selective attention to the relevant dimension failed (Fitousi, 2016; Fitousi & Wenger, 2013). The magnitude of GI indicates the level of dimensional interaction or cross-talk between the pertinent dimensions (Melara & Marks, 1990a, 1990b). Dimensions that produce substantial amounts of GI are dubbed *integral* dimension, whereas dimensions with which no GI is observed are called *separable* dimension (Garner, 1974, 1976; Garner & Felfoldy, 1970).

Integral Versus Separable Dimensions

The distinction between integral and separable dimensions entails qualitative differences in the ability of observers to allocate attention selectively. With integral dimensions, selective attention to the relevant dimension fails and GI surfaces; whereas with pairs of separable dimensions selective attention to the criterial dimension is

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good, and no GI is recorded (Algom & Fitousi, 2016). Objects that are composed of integral dimensions are processed in their totality as a Gestalt or a unified whole (Foard & Kemler, 1984). In contrast, objects that are composed of separable dimensions can be decomposed easily (Algom et al., 1996). With both types of dimensions, the constituting attributes are accessible to perception (Melara & Marks, 1990b). However, the primary difference is the facility with which the dimensions can be teased apart (Melara & Algom, 2003). Importantly, the distinction does not necessarily apply to objects. GI is the gold-standard measure of the level of correspondence or cross-talk between any pair of dimensions, whether they are part of the same object, or arrive from different sense modalities or spatial locations. For example, the Garner paradigm has been tested with cross-modality dimensions such as the spatial location of a dot (top vs. bottom) and a sounded pitch (high vs. low) (Melara & Marks, 1990a). In a matter of fact, a sizable portion of the Garner literature does not involve objects (Fitousi, 2016; Fitousi & Algom, 2006). Examples of integral dimensions are hue and saturation (Garner, 1974), saturation and brightness (Garner & Felfoldy, 1970), loudness and timber (Melara & Marks, 1990a), and facial identity and emotion (Fitousi & Wenger, 2013). Examples of separable dimensions are the size of a circle and the angle of a diameter (Shepard, 1964) and color and shape (Garner, 1974).

It is important to note that the integrality-separability distinction is supported by various *converging operations* (Garner et al., 1956; Von Der Heide et al., 2018), of which GI is only one (Garner, 1974). Garner himself noted that relying on a single operational definition as the sole arbiter of integrality is susceptible to circular reasoning. Therefore, he (Garner, 1974) and others (Fific et al., 2008; Hyman & Well, 1967; Shepard, 1964) have developed a series of converging procedures and theoretical approaches to the measurement of dimensional interaction. For example, the complete Garner paradigm consists of a third condition, *Correlation*, where values of the irrelevant dimension also vary from trial-to-trial. Under this condition, they vary in correspondence with the values of relevant dimension (e.g., a circle always appears in red and a square in green). When performance in the correlation block is better than in baseline, we term this advantage *redundancy gain* (Biederman & Checkosky, 1970; Garner, 1976). Integral dimensions often exhibit redundancy gains, while separable dimensions do not. However, the correlation condition has been criticized as an inadequate measure of separability, because it confounds decision bias with perception (Ashby & Maddox, 1994). Therefore, the current effort will focus on GI.

The integrality-separability distinction gained support from other converging operations, such as similarity scaling (Melara, 1992; Nosofsky, 1986). When participants are asked to judge the similarity of multidimensional objects, it is often found that integral dimensions "lead to a Euclidean metric in direct distance scaling" (Garner & Felfoldy, 1970, p. 238). In contrast, separable dimensions "yield City-block metric in direct-distance similarity scaling" (Monahan & Lockhead, 1977, p. 95). Another set of converging operations is provided by System Factorial Technology (SFT, Townsend & Nozawa, 1995). SFT is a stochastic theory augmented by a set of allied tools. These are designed to uncover fundamental aspects of information processing, such as architecture (serial vs. parallel), and stopping rule (i.e., exhaustive, self-terminating). In a landmark study, Fific et al. (2008) have harnessed the SFT methodology to show that separable dimensions are processed according to parallel or serial architectures, whereas integral dimensions are processed according to a coactive architecture (see also, Liu et al., 2022). Other sources of support in the integral/separable distinction come from General Recognition Theory (GRT, Ashby & Townsend, 1986), which is a multidimensional elaboration of signal detection theory (Green & Swets, 1966). GRT can dissociate between various types of dimensional and featural interactions (Ashby et al., 1996; Fitousi, 2018).

What Causes Interference?

GI requires a theoretical explanation that goes beyond the observed empirical phenomenon. As Garner (1974) himself noted: "Simply to state that the increased processing time obtained with irrelevant integral dimensions is due to a failure of selective attention is only to restate the experimental result" (p. 140). In response to this challenge, two explanations have been proposed to account for the emergence of GI. Both emanate from built-in key differences between filtering and baseline blocks. First, note that the number of variant dimensions in baseline is different from that in filtering. In baseline, the irrelevant dimension is held constant from trial-to-trial, whereas in filtering it changes in an orthogonal fashion. This difference has often been pointed out by proponents of the *irrelevant variation* account (Felfoldy, 1974; Garner, 1974). According to this account, greater uncertainty with respect to the irrelevant dimension slows down performance in filtering. Second, note that the number of stimuli in filtering and in baseline is different. Participants have to classify four stimuli in filtering, but only two in baseline. This mismatch in the number of stimuli has given currency to the *stimulus uncertainty* account (Ashby & Maddox, 1994). According to this explanation, GI is the result of larger stimulus uncertainty in filtering than in baseline. Which of the two accounts—irrelevant variation or stimulus uncertainty—is responsible for GI?

Dissociating between irrelevant variation and stimulus uncertainty in the Garner paradigm has proven to be a daunting task (Garner, 1974). A study by Burns (2016) addressed this challenge by introducing a new condition—*Correlated Filtering*—utilizing three stimulus dimensions (instead of two) to hold the number of stimuli constant, while introducing variation along a second irrelevant dimension. GI increased in this block, an outcome that supported the irrelevant variation account. In yet another study, aiming at pinpointing the nature of irrelevant variation itself, Niv et al. (2022) have tested the possibility that irrelevant variation is confounded with information content. These researchers astutely observed that because the irrelevant dimension in baseline is held constant (at one value), it provides no information to the participant; whereas in the filtering block, irrelevant variation creates uncertainty with respect to the irrelevant dimensions, and therefore it conveys information to the observer. According to Niv et al. (2022), this critical difference in information load is responsible for GI. To test for this prediction, they have added a "roving baseline" condition in which the value of the irrelevant dimension on trial n was completely determined by its value on trial $n-1$ (as is the case in regular baseline), but the values varied from trial-to-trial (as in filtering). A sizable GI emerged in this roving baseline, entailing that it is an irrelevant variation and not informational content that generates GI. These studies are important because they provide strong support in the notion that GI is generated by irrelevant variation and not due to stimulus uncertainty.

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However, irrelevant variation is not universally accepted by all researchers as the root cause of GI. In a matter of fact, the majority of computational models developed to account for GI put the blame on the other theoretical alternative—stimulus uncertainty. In one of the earliest computational models of GI, Ashby and Maddox (1994) have applied GRT (Ashby & Townsend, 1986) along with the RT-distance hypothesis (Murdock, 1985), to enable RT predictions. According to this model, the greater number of stimuli in filtering increases stimuli uncertainty, which in return, increases the variances of perceptual distributions, and slows down responses. Another prominent model is the *tectonic theory* (Melara & Algom, 2003). This model identifies the factor of dimensional uncertainty, which practically can be reduced to the number of stimuli, as responsible for GI. A third formal model is the Exemplar Based Random Walk (EBRW, Nosofsky & Palmeri, 1997). This model assumes that stimuli are represented as exemplars in memory and compete for retrieval according to a similarity-based random-walk process. EBRW predicts that the larger the number of stimuli (and therefore exemplars) in filtering the greater is the impedance to performance due to decreasing in strength of memory trace for each particular exemplar. However, the EBRW also assumes that the degree of similarity among stimuli also affects performance, a factor that is not solely dependent on the number of stimuli. In sum, while empirical studies support the irrelevant variation account, computational theories mostly assume that the number of stimuli is the cause of GI. Another plausible mechanism that can explain GI is sequential effects at the microlevel. This type of mechanism is discussed next.

Sequential Effects: The Key to Interference?

How does irrelevant variation or/and stimulus uncertainty cause interference? A potentially promising venture to explore this caveat is by looking at trial-to-trial sequential effects. Such effects have been shown to play a prominent role in diverse cognitive phenomena (Bertelson, 1961; Jones et al., 2013; Lockhead & King, 1983; Stewart et al., 2002). In the context of the Garner paradigm, Felfoldy (1974) examined successive trials in which both stimulus and response repeated (S & R repeat), both changed (S & R change), or in which stimulus changed and response repeated (S changes & R repeats). It is clear that the baseline block consists of only two of the three types of sequential trials that exist in the filtering block (i.e., S & R repeat or S & R change). This difference is crucial, because it may allow participants to engage more often in a "quick-check strategy" in baseline than in filtering blocks. In this simple tactic, participants change their response only if the now-presented stimulus is different from the previous stimulus; otherwise, they repeat their previous response. This strategy is also known as "bypass-rule" (Krueger & Shapiro, 1981), which states that processing is facilitated when both the stimulus and response are identical across consecutive trials and when both the stimulus and response are different across consecutive trials (Rabbitt, 1992).

Garner (1974) strongly endorsed this explanation. He was convinced that the interference bearing his name is the result of sequential effects at the microlevel: "… the overall increase in RTs from baseline to filtering with integral stimuli is largely due to microprocesses occurring with respect to different sequences of stimuli and responses" (p. 144). Garner pointed out two types of sequential effects. First, the quick check strategy—which he argued is very much like the reaction to a "blob" (Lockhead, 1972), and is evidently possible only with integral objects. The second is a further interference that is generated when the stimuli change on successive trials even when the response remains constant (i.e., S changes & R repeats), a type of sequence that is unique to filtering. A third type of sequential effects are repetition effects in which faster RTs are observed when the same stimulus repeats (Felfoldy, 1974). Such repetition effects have been first observed in the RT literature by Hyman (1953). Subsequent studies have shown that the magnitude of the repetition effect decreased as a function of the ensuing lag between consecutive stimuli (inter-stimulus-interval [ISI]) (Bertelson, 1961; Bertelson & Renkin, 1966; Hale, 1967; Smith, 1968).

The majority of investigators who studied GI (Felfoldy, 1974; Garner, 1974; Little et al., 2016) have assumed that sequential effects are generated by changes/repeats of entire stimuli and/or responses. However, an alternative conceptualization of sequential effects in Garner's paradigm has been proposed by Huettel and Lockhead (1999) and by Dyson and Quinlan (2010). Instead of looking at stimulus as whole entities, these researchers examined changes/repeats of specific *features* of the relevant and irrelevant dimensions. So, for example, assume that the letters *A* and *B* denote the relevant and irrelevant dimensions, respectively, and the subscripts 1 and 2 denote the feature levels of a given dimension. Now, if the stimulus A_1B_1 changes to A_2B_2 then levels of both relevant and irrelevant dimensions undergo a change (CC). If stimulus A_1B_1 is followed by A_2B_1 then only the level of the relevant dimension changes and the level of irrelevant dimension remains the same (CR). In this way, four types of sequential trials can be identified in filtering (RR, RC, CR, and CC), and two in baseline (RR and CR). Table A1 in the "Appendix" presents an exhaustive list of all the possible sequences in baseline and filtering blocks, along with their taxonomy according to this type of analysis. Dyson and Quinlan (2010) have used this approach to show that GI can be decomposed into several microlevel components:

$$\begin{aligned} \mathrm{GI} = {} & [E(\mathrm{RT}|\mathrm{RR};\ \mathrm{filtering}) + E(\mathrm{RT}|\mathrm{RC};\ \mathrm{filtering}) \\ & + E(\mathrm{RT}|\mathrm{CR};\ \mathrm{filtering}) + E(\mathrm{RT}|\mathrm{CC};\ \mathrm{filtering})]/4 \\ & - [E(\mathrm{RT}|\mathrm{RR};\ \mathrm{baseline}) + E(\mathrm{RT}|\mathrm{CR};\ \mathrm{baseline})]/2. \end{aligned} \tag{3}$$

In this way, it becomes apparent that filtering adds two sources of *irrelevant variation* (i.e., RC and CC) that do not exist in baseline. But these components also add an increase in *stimulus uncertainty*. Dyson and Quinlan (2010) derived additional macrolevel equations to dissociate the contributions of the two sources to GI. Empirical testing of those equations revealed that the contributions of irrelevant variation and stimulus uncertainty vary with the type of dimensions tested. Notwithstanding their remarkable originality, these results do not provide a conclusive answer on the origin of GI, nor do they explain how GI emerges with integral but not separable dimensions. They do, however, point to the importance of sequential effects in GI.

Sequential Effects and Garner Interference: Problems and Challenges

It is difficult to overemphasize the importance of sequential effects in generating GI. It is likely that GI can be completely accounted for by such effects (Dyson & Quinlan, 2010; Felfoldy,

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1974; Huettel & Lockhead, 1999). However, there are several challenges that have not been addressed yet. First, as noted by Burns (2016) the same structure of sequential trials is present with both *separable* and *integral* dimensions. Still, GI is observed only with integral dimensions. Why is it so? If one cannot explain the empirical differences between integral and separable dimensions, then the entire Garnerian edifice and associated fields of study are jeopardized (for discussion of this point see, Niv et al., 2022). Second, GI is probably generated not by one, but several types of sequential effects (Felfoldy, 1974; Garner, 1974). Thus, one should seek for a unified framework, one that can explain all of these effects as resulting from a single process or mechanism. Currently, no such account is available. Third, a viable theoretical model should explain how the magnitude of GI is determined by the strength of sequential effects. What are the mechanisms that can explain (if exists) such quantitative relations? Fourth, the proposed accounts—stimulus uncertainty and irrelevant variation—are not really psychological mechanisms, but rather contributing factors, that likely affect a deeper underlying process. All of these caveats should be addressed within a complete theoretical framework. The current study proposes such a framework.

Binding Mechanisms in Sequential Processing

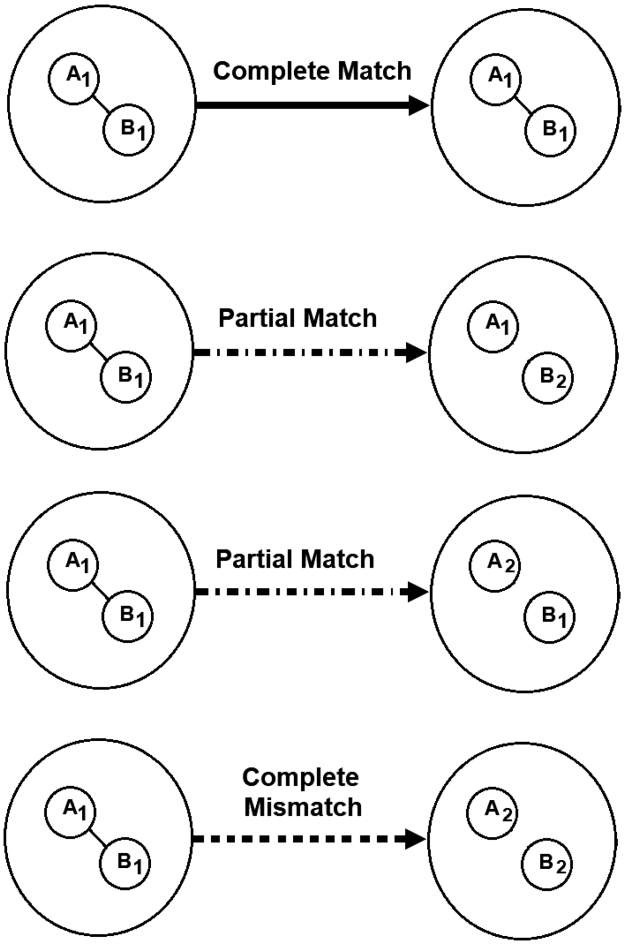
The present study offers an alternative approach to the reviewed findings on sequential effects (Dyson & Quinlan, 2010; Felfoldy, 1974). Here, GI and dimensional interaction are attributed to a feature integration mechanism that operates on features across consecutive trials. The theoretical basis for such processes has been outlined in a seminal study by Kahneman et al. (1992) who have proposed that the codes representing the features of the same object are integrated in a temporary episodic memory they called "*object files*." In support of their conjecture, Kahneman et al. (1992) showed that participants respond faster and more accurately if the identity of a visual object is repeated, but only in cases that preserve the relations between identity and spatial location, which suggests that the previous encounter has created an identity-location binding trace. Hommel's (1998) extended this work in several important ways. First, he showed that "object files" can be created through bindings between features such as color and shape. Thus, *complete repetitions* of both features of a stimulus (e.g., red circle → red circle) or *complete alterations* (e.g., red circle → green square) results in faster RTs and lower accuracy rates than *partial repetitions*, in which one stimulus feature is repeated, but the other feature alternates (e.g., red circle → red square). These *partial repetition costs* strongly support the notion of "object files"—temporary episodic bindings of features. If a partial repetition retrieves a previous object file that associates the repeated feature with another feature that is not part of the present stimulus, this introduces feature competition. Competition increases reaction time, simply because more or longer analysis of the currently present stimulus is necessary to override the competing activation.

Second, Hommel (1998, 2004) showed that spatial location is not important for the creation of "object files," to the extent that alternating or repeating the location of a stimulus does not affect binding across, for example, its color or shape. Figure 1 provides an illustration of the process that governs feature bindings. In addition, Colzato et al. (2006) have shown that feature-binding is strong for highly learned objects (e.g., red strawberry and yellow banana), but fragile for arbitrary conjunctions (e.g., shape and color). They interpret these results as reflecting two binding mechanisms, a long-term mechanism that supports binding with familiar feature combinations, and a short-term integration process that is responsible for ad-hoc binding (Hommel & Colzato, 2009).

Another contribution of Hommel's feature-binding approach (Hommel's, 1998; Hommel, 2004) is in demonstrating the presence of similar bindings effects between response and stimulus features, which he termed "*event files*." Thus, complete repetitions of response and stimulus feature ($R1 + S1 \rightarrow R1 + S1$) or complete alternation ($R1 + S1 \rightarrow R2 + S2$) results in more efficient performance than

Figure 1

Four Possible Combinations of Features Conjunctions Resulting From Feature Repetition or Alteration of Features in a Succession of Two Stimuli Composed of Two Dimensions (A vs. B) That Vary Along Two Levels of Features (1 vs. 2)



Note. The left column represents the preceding stimulus, while the right column depicts the current stimulus. The main idea behind "object files" is that the features of the first stimulus are integrated into a single unit. Activating one feature code by the second stimulus also activates the other feature code it is bound with. As a result, a partial match leads to the activation of wrong and therefore misleading stimulus features. "Event-files" can be understood in a similar fashion, with responses (R_1, R_2) replacing the role of features in the relevant dimension.

with partial repetitions ($R1 + S1 \rightarrow R1 + S2$).[1] Extensive research on the temporal (Hommel & Colzato, 2004) and memorial aspects (Fitousi, 2018; Frings et al., 2013; Frings & Rothermund, 2011; Hommel, 2005; Hommel & Frings, 2020; Moeller et al., 2016) of "object files" and "event files" has been conducted. The approach has proven immensely productive in explaining a myriad of cognitive phenomena, including sequential effects in conflict tasks (Hommel et al., 2004), and the perception of complex objects such as faces (Fitousi, 2017a, 2017b; Keizer et al., 2008). In the current work, I demonstrate how similar binding concepts can be harnessed to explain the emergence of GI and related phenomena.

A Sequential Feature Binding Account of Garner Interference

The episodic sequential binding account of GI builds on: (a) earlier classic concepts of "object files" and "event files" (Hommel's, 1998; Kahneman et al., 1992), (b) the ideas of Dyson and Quinlan (2010) concerning the decomposition of the GI to sequential components (Huettel & Lockhead, 1999), and (c) previous notions of stimulus uncertainty and irrelevant variation. According to this account, GI emerges due to the formation of object files or event files. A main assumption of the present account is that integral dimensions create a "blob" (Garner, 1974) or a configural object, which exerts stronger binding costs than those induced by separable dimensions. In particular, on each trial the participant encodes the featural values of the stimulus along the two dimensions, preserving and retaining them in her episodic memory as an "object files" or an "event file" (Hommel's, 1998; Kahneman et al., 1992). Thus, on trial $n-1$ the participant creates an episodic unit that consists of a unitary binding of the two constituting levels of the relevant and irrelevant dimensions. When the new "object file" is a complete replica of the previous one (complete repetition of features), or a new one (complete alteration of the features), then retrieval processes are fast and efficient. However, when the features of the new stimulus represent a partial repetition of those in the previous stimulus (i.e., one feature is repeated but the other changes) feature competition takes place which prolongs RTs and leads to more errors. It is predicted that the emergence of GI with integral dimensions will be associated with stronger binding, and therefore, with larger *partial repetition costs* than those associated with separable dimensions. It is also assumed that partial repetition costs reflect a true binding process only in filtering and not in baseline, as explained next.

These ideas can be cast in a more quantitative manner by decomposing the GI into various types of sequential trials (Dyson & Quinlan, 2010; Huettel & Lockhead, 1999), and showing how partial repetition costs that result from these components are numerically related to the magnitude of GI. First, an estimator of the amount of partial repetition costs in filtering can be given using the *mean interaction contrast* (MIC) term based on the repetition or alteration of two dimensions with two features (see also Hommel, 2005):

$$\begin{aligned}\text{cost(filtering)} = [&E(\text{RT}|\text{RC; filtering}) + E(\text{RT}|\text{CR; filtering})] \\ &- \left[E(\text{RT}|\text{CC; filtering}) + E(\text{RT}|\text{RR; filtering})\right].\end{aligned} \tag{4}$$

This MIC quantifies the strength of binding as the difference in performance between the partial repetition trials (RC, CR) and the complete repetition/alteration trials (RR and CC). Similarly, an estimator for the amount partial repetition costs in baseline can be written as a difference between the partial repetition (CR) and complete repetition (RR) sequences:

$$\begin{aligned}\text{cost(baseline)} = [&E(\text{RT}|\text{CR; baseline}) \\ &- E(\text{RT}|\text{RR; baseline}).\end{aligned} \tag{5}$$

A crucial assumption in the theory is that only in filtering do partial repetition costs reflect a true binding process, whereas in baseline such costs gauge a unidimensional repetition effect. Returning to Equation 3 derived by Dyson and Quinlan (2010), we note that filtering consists of four types of sequential trials (RR, RC, CR, and CC), whereas baseline consist of two types (RR and CR). Thus, filtering adds two sources of *irrelevant variation* (i.e., RC and CC) that do not exist in baseline, which also add an increase in *stimulus uncertainty*. Note that this is due to the fact that in baseline, the irrelevant dimension is held fixed at a single value (feature), whereas in filtering the irrelevant dimension varies orthogonally from trial-to-trial, and, therefore, the features change in a manner that constitutes a real dimension. In terms of information theory (Garner, 1962; Hick, 1952; Shannon, 1948), the irrelevant dimension in filtering transmits information due to the presence of uncertainty in its possible values. In contradistinction, the irrelevant dimension in baseline does not transmit any information, because there is full certainty with respect to its constituting values (it is always the same repeating value). It is therefore likely, that due to the absence of irrelevant variability in the baseline block, the irrelevant dimension is not perceived as a dimension at all ! According to this theoretical interpretation, no attention is allocated to the irrelevant single feature in baseline and it is therefore not integrated with the features of the relevant dimension. This entails that the difference between CR and RR in baseline does not reflect binding, but rather a simple (uni-dimensional) repetition effect [$E(\text{RT} \mid \text{CX}) - E(\text{RT} \mid \text{RX})$], where X is the fixed irrelevant feature. Similar arguments have been made by Garner (1974, 1976). This conjecture was supported by both formal derivations and empirical results, as explained next.

By combining Equations 2–4, it becomes apparent (see "Appendix B") that GI also amounts to:

$$\begin{aligned}\text{GI} = &E(\text{RT}|\text{cost; filtering})/4 - E(\text{RT}|\text{Baseline})/2 \\ &+ [E(\text{RT}|\text{RR; filtering}) + E(\text{RT}|\text{CC; filtering})]/2.\end{aligned} \tag{6}$$

Thus, for GI to emerge (GI > 0), it can be readily shown that partial repetition costs in filtering should exceed the sum of two times performance in baseline minus twice the benefits of joint repetitions and joint alterations in filtering:

$$\begin{aligned}E(\text{RT}|\text{cost; filtering}) > &2 \times E(\text{RT}|\text{Baseline}) \\ &-2 \times [E(\text{RT}|\text{RR; filtering}) \\ &+ E(\text{RT}|\text{CC; filtering})].\end{aligned} \tag{7}$$

[1] Note that this phenomenon is comparable to the bypass-rule (Krueger & Shapiro, 1981), according to which processing is facilitated when both the stimulus and response are identical across consecutive trials and when both the stimulus and response are different across consecutive trials (Rabbitt, 1992).

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This inequation reveals a fundamental property of the GI—its degree (and very existence) depends on the strength of feature binding in filtering. This conclusion is also consistent with the idea that binding cannot take place in baseline, because the irrelevant dimension is held fixed at one value, and does not function as a dimension at all. It can also be noted that partial repetition costs get larger as the joint alteration and joint repetition trials get faster, because they are subtracted from performance in baseline. Taken together, the derivations reveal that, GI occurs only when binding in filtering exceeds a given magnitude. Most importantly, binding can potentially occur with separable dimensions (in a matter of fact Hommel's (1998) original studies deployed color and shape which are often considered to be separable dimensions), but it is expected to be much larger with integral than separable dimensions. One implication of the theory is that, with respect to feature binding, integral and separable dimensions differ quantitatively not qualitatively. The difference emanates from differential amounts of partial repetition costs in filtering. Another insight is that the magnitude of GI is related parametrically to the strength of binding in filtering. The theory relates two otherwise remote theoretical notions—integral or holistic processing (Lockhead, 1972) on the one hand, and feature binding (Kahneman et al., 1992)—on the other hand, by a single explanatory mechanism.

It should be acknowledged though, that in the Garner paradigm the relevant features (e.g., red vs. green) to which the observer attends, are confounded with their mapped responses (e.g., left vs. right). Thus, a change in the relevant dimension (feature) also implies a change in the required response (see Table A1 in "Appendix A"). It is therefore possible that binding takes place between the codes of the relevant response and irrelevant feature (response–stimulus binding), rather than between the codes of relevant and irrelevant features (stimulus–stimulus binding). In that case, GI is generated by the formation of the so-called "event-files" rather than "object-files" (Hommel's, 1998). A response–stimulus binding entails a different species of integration than that of stimulus–stimulus binding (Fitousi, 2017b); but in the traditional Garner task, these two types of integration are indistinguishable.[2]

Predictions and Research Strategy

The current study proposes a new theory of dimensional interaction. The basic idea is that the level of cross-talk between dimensions is driven by the degree of feature integration between relevant and irrelevant features (object-files), or between relevant responses and irrelevant features (event-files). The theory predicts that integral and separable dimensions differ in the strength of binding they induce among their features. Stronger feature binding contributes to higher levels of cross-talk between the dimensions. Thus, the theory predicts larger bindings costs for integral than separable dimensions. In particular, partitioning of GI onto its bindings elements leads to a novel theory of the effect. According to this account, GI is generated through a binding mechanism that operates across consecutive trials. The theory postulates automatic encoding of relevant and irrelevant features in working memory (Logan, 1988), which in turn affects processing of subsequent stimulus. Thus, when a stimulus is presented on trial $n-1$ participants encode its features from both dimensions and retain them in working memory as an "object file" or an "event file." On trial n participants have to process another conjunction of features. If this conjunction is a replica of the already stored "object file" ("event file"), or consists of a completely new set of features, then binding is easy. However, if this conjunction shares one feature with the stored "object file" ("event file"), then it interferes with the retrieval of a new object file (= partial repetition cost) due to competition (Hommel's, 1998). In addition, binding likely occurs only in the filtering blocks and not in baseline. The reason being that in baseline the irrelevant dimension is held fixed at one feature value. As a result, it is likely that observers do not perceive it as a "dimension" at all (see e.g., Garner, 1974). Thus, partial repetition costs in baseline reflect only a repetition effect of a single (e.g., relevant) dimension.

Indeed, decomposition of GI onto its binding elements revealed that it is solely affected by feature binding in filtering and not in baseline; and that GI can be obtained only when the strength of binding in filtering exceeds a given magnitude (see Equation 7). In addition, the ISI, namely, the time lag ensuing between consecutive stimuli, has been shown to exert significant influence on the magnitude, and very existence of sequential effects (Bertelson, 1961; Bertelson & Renkin, 1966; Hale, 1967). For example, it was found that the difference in RT between repetition and nonrepetition trials increases for a given ISI, or remains constant as ISI increases. The latter occurs when either the number of stimulus alternatives gets larger (Hyman & Umiltà, 1969; Smith, 1968) or the probability of repetition increases (Bertelson, 1961; Hyman, 1953; Hyman & Umiltà, 1969). Because the current study examines sequential effects with different number of stimulus alternatives in baseline and filtering, it is of theoretical importance to examine the potential influence of ISI. Moreover, a previous study (Felfoldy, 1974) revealed that the magnitude of GI itself decreased with ISI. One plausible explanation for this phenomenon is that the strength of feature binding weakens with time. In the Garner paradigm, a change in the relevant feature also implies a change in response. This entails that binding can take place either due to the formation of object files (integration of feature codes) or due to the formation of event files (integration of response codes and feature codes). While previous studies (Hommel, 2005; Hommel & Colzato, 2004) provided very little evidence for the decline of object files strength with time, Hommel and Frings (2020) did show a strong effect of ISI on the strength of event files.

The question of interest is therefore whether separable and integral dimensions differ with respect to the amount of binding they induce in the Garner paradigm. In Experiments 1 and 2, I have harnessed two pairs of integral dimensions (chroma and value, and width and height of rectangles). In Experiment 3, I have used a pair of separable dimensions (circle's size and angle of a diameter). In all experiments, I have manipulated the ISI between consecutive trials. The theory predicts that: (a) with integral dimensions (Experiments 1 and 2) the magnitude of GI will decrease as a function of ISI, (b) partial repetition costs will be correlated with the magnitude of GI in

[2] To dissociate the contribution of response from that of stimulus, a Reviewer proposed to replace the traditional two-alternative choice task with a four-alternatives choice task. This might be problematic, because such a response mapping alters the Garner task in a way that diverts it from its original intent. The Garner paradigm is a classic selective attention task. By asking the participant to respond with a four-alternative choice, the observer, of necessity, is required to allocate attention to the irrelevant dimension. Under this response regime, the task becomes a divided attention task, which is an anathema to the original purpose of the Garner paradigm (for discussion see Algom et al., 2022).

filtering, but not in baseline, and (c) partial repetition costs will decrease with ISI. The predictions for separable dimensions (Experiment 3) are as follows: (a) no GI or very small amounts of GI are expected, (b) partial repetition costs will be significantly smaller compared to those observed with integral dimensions, (c) partial repetition costs will not be correlated with the magnitude of GI.

Experiment 1

Munsell dimensions of value and chroma were tested. These are considered to be classic integral dimensions (Garner & Felfoldy, 1970), and are well known to produce robust GI. In this experiment, two independent groups of participants were asked to judge chroma (value) while ignoring irrelevant variation on value (chroma). In addition, two ISI levels of short (82 ms) and long (1,080 ms) lags were used.

Method

Participants

A total of 46 participants from Ariel University took part in the experiment. They consisted of two independent groups of 22 ($F = 18$, $M = 4$, $M_{age} = 23.34$), and 24 ($F = 20$, $M = 4$, $M_{age} = 22.56$) young adults. The first group performed in judgments of chroma, while the second group in classifications of value. All had normal or corrected-to-normal vision. All gave their signed consent.

The "pwr" package (Champely et al., 2018) in the statistical software R (R Core Team, 2017) was used to compute the sample sizes. The power was set to 0.80 and the significance level was set to 0.05. Based on previous studies (Fitousi, 2021; Fitousi & Algom, 2020) the effect size of the GI was considered to be large (0.50). The minimum N required is 13.

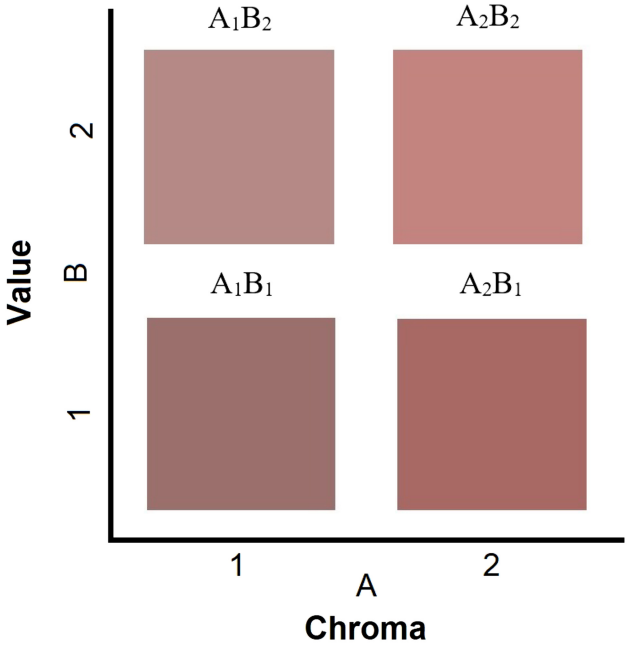
Stimuli and Apparatus

The stimuli consisted of four color rectangles of approximately 5 cm × 5 cm. Viewed from a distance of 56 cm the stimuli subtended 5.09° × 5.09° visual angle. The stimuli were generated according to a factorial intersection of two Munsell dimensions of chroma and value. The hue of the rectangles was 5R. The two levels of chroma were four or six and the two levels of value were five or six. These levels are the ones deployed by Garner and Felfoldy (1970) in their Experiment 1. The four resulting stimuli appear in Figure 2.

Procedure and Design

Participants sat in front of a desktop computer in a dimly lit room. Stimuli were color images presented over white background at the center of the screen. On each trial, one stimulus was randomly chosen by the computer and presented until the response. After the participant responded, the stimulus was replaced by a blank screen. One group of participants judged chroma and ignored value; the other group classified the stimuli along the value dimension and ignored chroma. The experiment consisted of two types of blocks. In filtering, both chroma and value varied in an orthogonal fashion, such that any one of the four stimuli could appear. For the group that judged chroma, the value in the baseline blocks was held constant at one level, confined to either five (Baseline 1) or six (Baseline 2). For the group that judged value, chroma in the baseline blocks was held constant, restricted to either four (Baseline 1) or six (Baseline 2).

Figure 2
The Stimuli Used in Experiment 1



Note. These colored squares were created by a factorial intersection of two Munsell levels of chroma (four and six) and two levels of value (five and six). Hue was held constant at 5R. Please note that the grayscale image in the print version does not convey the essential aspects of the stimuli. For their veridical appearance, please consult the online color version of this figure.

A filtering block consisted of 56 trials, while a baseline block consisted of 28 trials. Because there were two types of baseline blocks, the number of trials in filtering and baseline was equal (i.e., 56). The trials in each block were randomized by the computer with the constraint that each stimulus appeared equally often. The filtering and two baseline blocks comprised a cycle, and were run together in random order. A cycle consisted of 112 trials. Overall, participants completed four cycles for a total of 448 trials. ISI was randomized across cycles, being short (82 ms) in two cycles, and long (1,080 ms) in the other two. A break of 2 min separated each block. Participants responded with either their right or left hand by pressing the button keys ("m" or "z"). Participants were asked to respond as fast and as accurate as they could. Before the experiment began, participants engaged in a short training session that included 120 trials. This training session was similar to the experiment, but unlike the experiment, provided participants with a trial-by-trial feedback with respect to their accuracy. This session was held to guarantee that participants understood what chroma and value are, and that they follow the response assignment rule.

Results

Data can be downloaded here (Fitousi, 2023). The structure of sequential trials in the two baseline blocks was identical, thus in agreement with the standard practice in Garner studies (Algom et al., 1996; Fitousi, 2020; Melara & Mounts, 1993) data from these two blocks were collapsed.

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RTs shorter than 150 ms or longer than 1,800 ms were removed from analyses. These amounted to 5.8% of the total number of trials in judgments of chroma and 1.8% of trials in judgments of value. Error trials (9.1% and 5.5% in the chroma and value judgments, respectively) were also removed. Two types of computations were performed on the same dataset. One analyzed the magnitude of GI as a function of ISI. It is expected that GI will decrease as a function of ISI. The second analysis followed the logic of episodic binding proposed by Hommel's (1998). Data were analyzed as a function of repetition versus alternation of relevant (e.g., chroma) and irrelevant dimensions (e.g., value) across successive trials, separately in baseline and filtering blocks. The binding approach predicts that chroma and value will be integrated across consecutive trials. Thus, faster RTs are expected when both chroma and value values repeat (RR, complete match) or when both chroma and value alternate (CC, complete change) in comparison to sequences in which either chroma change and value repeats (CR, partial match) or chroma repeats and value changes (RC, partial match). The level of integration was estimated for short and long ISIs, expecting integration to decrease with ISI.

Garner Interference

Across-Tasks

I first tested for the presence of GI and its possible modulation by ISI across tasks (chroma, value). A three-way mixed-design ANOVA with Task (chroma, value) × Block (filtering, baseline) × ISI (82 ms, 1,080 ms) was conducted on mean RTs. Task served as a between-participant factor, whereas Block and ISI served as within-participant factors. This analysis revealed a main effect of Block, $F[1, 132] = 75.93$, MSE = 270, 279, $\eta_p^2 = 0.37$, $p < .001$, such that performance in filtering was slower than that in baseline. This GI entails that, irrespective of task, variation on the irrelevant dimension exacted a toll on selective attention to the criterial dimension. This finding supports the notion that chroma and value are integral dimensions, and that they produce symmetric GI (Garner & Felfoldy, 1970). That is, judgments of chroma are interfered by irrelevant variation on value; and similarly, classifications of value are compromised by irrelevant variation on chroma. Moreover, GI was modulated by ISI, $F[1, 132] = 7.34$, MSE = 26, 149, $\eta_p^2 = 0.05$, $p < .01$, such that larger GI (101 ms) was documented in short, $t[132] = 8.07$, $p < .0001$ than in long (53 ms), $t[132] = 4.24$, $p < .0001$ ISIs. This finding is consistent with an earlier result (Felfoldy, 1974). One goal of the current study is to explain this effect in terms of binding. In addition, an ISI × Block interaction was also significant, $F[1,132] = 8.02$,MSE = 28,563, $\eta_p^2 = 0.08$, $p < .01$, but was not subjected to further analysis due to lack of theoretical interest. No other effect was significant. Comparable analyses on error rates revealed significant main effect of Block, $F[1, 132] = 15.25$, MSE = 0.038, $\eta_p^2 = 0.10$, $p < .001$, with more errors made in filtering than in baseline, and supporting yet again the presence of GI. The first ANOVA suggests that the patterns observed are not modulated by the dimension to which participants attended. In other words, there is no qualitative difference in the observed patterns across tasks. However, because the current theory offers also a quantitative account of the magnitude of GI, it is of theoretical importance to examine how these effects manifest separately in judgments of chroma and value.

Classification by Chroma

The top panel of Figure 3 gives mean RTs and mean error rates in judgments of chroma. A two-way ANOVA with Block (filtering, baseline) and ISI (82 ms, 1,080 ms) as factors revealed that the effects of Block, $F[1, 21] = 88.03$, MSE = 11, 460, $\eta_p^2 = 0.80$, $p < .001$, ISI, $F[1, 21] = 5.18$, MSE = 24, 474, $\eta_p^2 = 0.19$, $p < .05$, and their interaction, $F[1, 21] = 10.61$, MSE = 6, 845, $\eta_p^2 = 0.33$, $p < .005$, were all significant. The interaction entails that GI was modulated by ISI. Planned comparisons confirmed that GI was much larger (148 ms) in short, $F[1, 21] = 88.22$, MSE = 86, 774, $\eta_p^2 = 0.80$, $p < .0001$, than (57 ms) in long, $F[1, 21] = 33.99$, MSE = 31, 531, $\eta_p^2 = 0.61$, $p < .005$, ISIs. This result is consistent with the episodic binding account, according to which reduction in strength of "object files" or "event files" in working memory should lead to reduction in the magnitude of GI. Comparable analyses on error rates revealed a main effect of Block, $F[1, 21] = 14.73$, MSE = 0.041, $\eta_p^2 = 0.41$, $p < .001$, such that participants made more errors in filtering than in baseline. GI was modulated by ISI, $F[1, 21] = 14.55$, MSE = 0.0064, $\eta_p^2 = 0.40$, $p < .005$. Planned comparisons confirmed that the interference was larger (6%) in short, $F[1, 21] = 19.82$, MSE = 0.040, $\eta_p^2 = 0.48$, $p < .0001$ than in long (2.6%), $F[1, 21] = 6.16$, MSE = 0.0075, $\eta_p^2 = 0.23$, $p < .05$, ISIs. Overall, these results support the idea that classification of chroma was interfered by irrelevant variation on value, and that this GI decreased as a function of ISI.

Classification by Value

The bottom panel of Figure 3 gives mean RTs and mean error rates in judgments of value. Judgments of value were slower in filtering than in baseline, $F[1, 23] = 46.63$, MSE = 109, 409, $\eta_p^2 = 0.66$, $p < .0001$. This main effect documents a sizable GI, which points to the breakdown of selective attention to value in the face of irrelevant variation from chroma. GI was modulated by ISI, as indicated by the significant two-way Block × ISI interaction, $F[1, 23] = 4.52$, MSE = 5, 717, $\eta_p^2 = 0.16$, $p < .05$. GI was larger (83 ms) in short ISI, $F[1,23] = 63.96$, MSE = 82,572, $\eta_p^2 = 0.73$, $p < .0001$, than in long (52 ms) ISI, $F[1, 23] = 14.05$, MSE = 32, 554, $\eta_p^2 = 0.37$, $p < .005$. A main effect of ISI was not significant, $F[1, 23] = 1.59$, MSE = 10, 277, $\eta_p^2 = 0.06$, $p = .21$. Comparable analyses on error rates revealed a GI, $F[1, 23] = 11.0$, MSE = 0.0005, $\eta_p^2 = 0.33$, $p < .005$. No other effect was significant. These analyses confirm that judgments of value were interfered by irrelevant variation on chroma, and that this GI was modulated by ISI.

Taken together, the results from the two tasks support the mutual integrality of chroma and value, and the shrinking of GI with ISI.

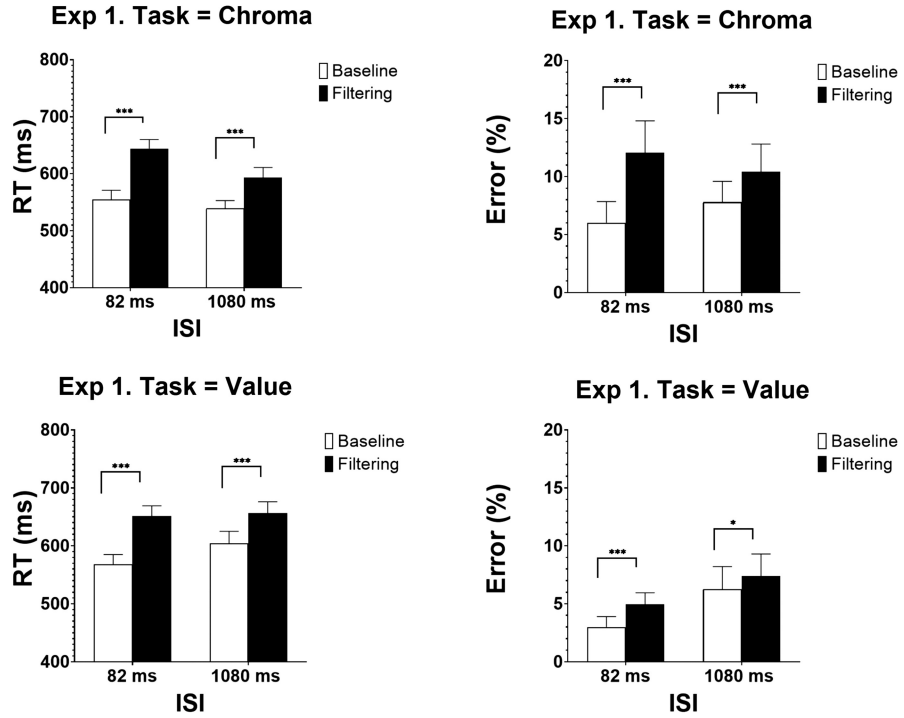
Partial Repetition Costs in Filtering

Across-Tasks

A four-way mixed-design ANOVA with Task (chroma, value) × Chroma (change, repeat) × Value (change, repeat) × ISI (82 ms, 1,080 ms) was conducted. Task served as a between-participant factor. In this analysis, a main effect of chroma, $F[1, 308] = 17.6$, MSE = 86, 545, $p < .001$, and a main effect of value, $F[1, 308] = 59.12$, MSE = 290, 665, $p < .001$, obtained. These are repetition effects, in which faster RTs are documented when

Figure 3

Experiment 1 (Chroma and Value): Mean RTs and Error Rates as a Function of Block (Baseline, Filtering) and ISI



Note. Error bars are standard error of the mean. RTs = response times; ISI = inter stimulus interval.
* $p < .05$. ** $p < .01$. *** $p < .0001$.

chroma (value) is repeated compared to when it is changed. But these priming effects are not indicative of binding. Most importantly, a Chroma × Value interaction, $F[1, 308] = 58.17$, MSE = 285, 973, $p < .001$ and its modulation by ISI, $F[1, 308] = 5.20$, MSE = 25, 606, $p < .05$, were found to be significant. The interaction implied faster RTs when both features repeated or changed together compared to when one feature repeated, but the other changed. These patterns provide support in the presence of a feature binding effect that is modulated by ISI. Further analyses were performed separately for chroma and value judgments.

Classification by Chroma

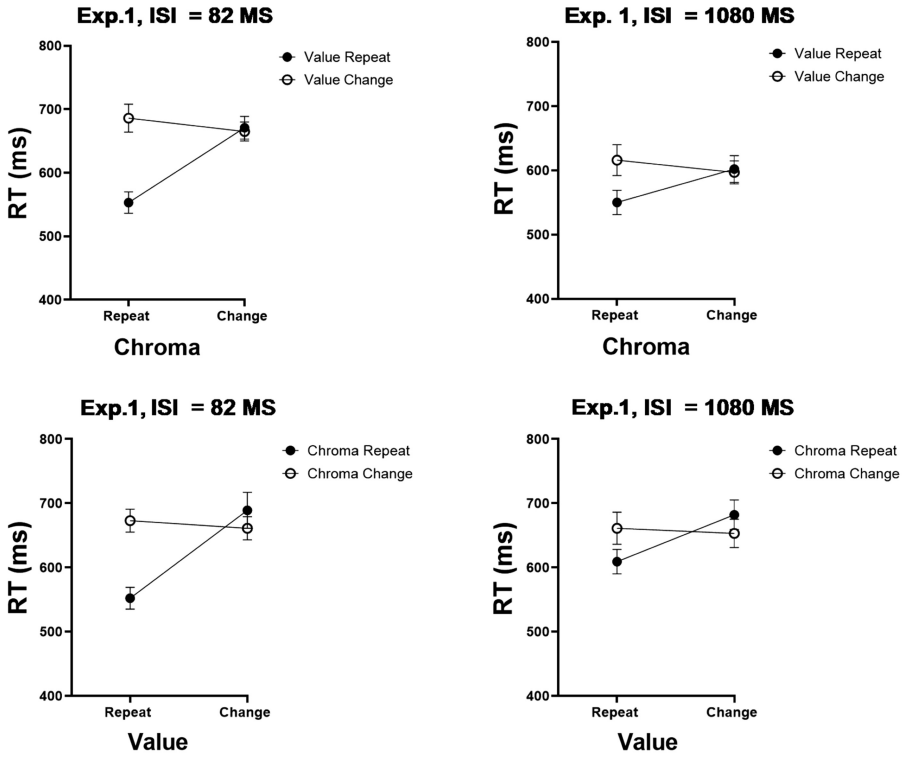
The top panel of Figure 4 gives mean RTs and error rates in this task. A three-way ANOVA with Chroma (change, repeat) × Value (change, repeat) × ISI (82 ms, 1,080 ms) was conducted. If indeed binding occurs across successive trials, then one should expect the two-way interaction of Chroma × Value to be significant, such that faster RTs are predicted when both relevant and irrelevant feature values repeat (RR) or alternate (CC) compared to trials in which one of them repeats but the other alternates (RC or CR). This pattern is also called *partial feature repetition cost*. Moreover, if this binding effect gets weaker with time, then it should be modulated by ISI. This will be reflected as a three-way interaction of Chroma × Value × ISI. A repetition effect for Value, $F[1, 21] = 48.08$, MSE = 97, 531, $\eta_p^2 = 0.63$, $p < .001$, was documented, such that participants responded faster when value was repeated (594 ms) than when it was altered (642 ms). A similar repetition effect was found for Chroma, $F[1, 21] = 15.13$, MSE = 46, 313, $\eta_p^2 = 0.41$, $p < .001$, with participants responding faster to repeated levels of chroma (602 ms) than to altered levels (634 ms). It should be noted again that these effects are not consequential for the observation of binding effects (Hommel's, 1998), and may or may not be present in binding processes. Binding is genuinely signaled by a two-way interaction.

Most importantly, a Chroma × Value interaction was found to be significant, $F[1, 21] = 11.04$, MSE = 119, 924, $\eta_p^2 = 0.34$, $p < .005$, capturing the expected pattern by which complete repetitions and alternations (RR = 552 ms and CC = 631 ms) were responded to faster than partial repetitions (RC = 652 ms, CR = 637 ms). This pattern entails that, on a given trial, chroma and value were integrated into an object-file or an event-file (see Figure 4). In addition, the effect of ISI was significant, $F[1, 21] = 11.04$, MSE = 119, 924, $\eta_p^2 = 0.34$, $p < .005$, suggesting that responses were faster following a long (592 ms) than a short (644 ms) ISI. More important is the finding that the Value × Chroma interaction was modulated by ISI, $F[1, 21] = 7.71$, MSE = 12, 505, $\eta_p^2 = 0.26$, $p < .05$, a result that points to stronger partial repetition costs in short than in long ISIs. The MIC, which serves as an estimator of binding strength (see Equation 4) was larger (MIC = 139 ms) in short than in long (MIC = 71 ms) ISI ($t[21] = 2.77$, $p < .01$). A pair of two-way interactions was significant: ISI × Chroma, $F[1, 21] = 5.20$,

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Figure 4

Experiment 1: Mean RTs in Filtering as a Function of Repetition and Alterations of Features of the Relevant and Irrelevant Dimension at Two Levels of ISI



Note. Error bars are standard error of the mean. RTs = response times; ISI = inter stimulus interval.

MSE = 11, 041, $\eta_p^2 = 0.55$, $p < .05$, and ISI × Value, $F[1, 21] = 26.64$, MSE = 11, 658, $\eta_p^2 = 0.55$, $p < .0001$, but these were not subjected to further analysis due to lack of theoretical interest. Comparable analyses on error rates provided evidence for binding in the form of a significant Chroma × Value interaction, $F[1, 21] = 6.24$, MSE = 0.0183, $\eta_p^2 = 0.22$, $p < .05$. Although binding was numerically larger (MRI = 6%) in short than in long (1.5%) ISI, the pattern was not significant ($t[21] = 1.47$, $p = .07$).

Classification by Value

The bottom panel of Figure 4 presents mean RTs and error rates in this task. A repetition effect was observed for the relevant dimension of Value, $F[1, 23] = 33.99$, MSE = 107, 446, $\eta_p^2 = 0.59$, $p < .001$, and for the irrelevant dimension of Chroma, $F[1, 23] = 8.0$, MSE = 39, 648, $\eta_p^2 = 0.25$, $p < .005$. The effect of ISI was not significant [$F < 1$]. Recall that repetition effects are not consequential for binding. They may or may not be documented along with binding effects (Fitousi, 2017a, 2017b). Most importantly, partial repetition cost was indicated by a significant Value × Chroma interaction, $F[1, 23] = 57.63$, MSE = 158, 884, $\eta_p^2 = 0.71$, $p < .001$, which documented the expected pattern by which complete repetitions and alternations (RR = 581 ms and CC = 657 ms) were responded to faster than partial repetitions (RC = 686 ms, CR = 667 ms). The partial repetition cost was modulated by ISI, as indicated by three-way interaction Value × Chroma × ISI, $F[1, 23] = 7.55$, MSE = 14, 426, $\eta_p^2 = 0.26$, $p < .05$. This interaction entailed stronger partial repetition costs in short than long ISIs. The estimator of partial repetition costs, which signals binding in filtering (see Equation 4) was larger (MIC = 149 ms) in short than in long (MIC = 80 ms) ISIs, $t[23] = 2.74$, $p < .01$. In addition, the ISI × Chroma, $F[1, 23] = 57.63$, MSE = 158, 884, $\eta_p^2 = 0.71$, $p < .0001$, and ISI × Value, $F[1, 23] = 6.20$, MSE = 11, 016, $\eta_p^2 = 0.21$, $p < .05$, interactions were significant, but not subjected to further analysis due to lack of theoretical interest. Comparable analyses on error rates revealed a main effect of the magnitude of partial repetition costs, $F[1, 23] = 8.94$, MSE = 0.016, $\eta_p^2 = 0.28$, $p < .005$. But this effect was not modulated by ISI [$F < 1$].

Taken collectively, these analyses support the presence of sequential binding in the filtering block for both judgments of chroma and value. The strength of binding was found to be modulated by ISI, such that larger partial repetition costs are observed with short than long ISI.

Repetition Effects in Baseline

Next, I assessed the presence of repetition effects in baseline using a three-way ANOVA with Task (chroma, value) × Trial (RR, CR) × ISI (short, long) as factors. The effect of Trial, $F[1, 132] = 34.86$, MSE = 130, 518, $p < .0001$, was significant, but it was not modulated by ISI [$F < 1$]. Separate two-way

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ANOVAs with Trial × ISI were performed for judgment of chroma and value. For chroma, a main effect of trial was only significant, $F[1, 21] = 5.2$, MSE = 11, 568, $\eta_p^2 = 0.18$, $p < .05$. Notably, the absence of Trial × ISI interaction [$F < 1$] suggests that the effect of repetition was comparable in short and long ISIs. Similarly, for judgment of value, a main effect of Trial, $F[1, 23] = 55.27$, MSE = 47, 358, $\eta_p^2 = 0.70$, $p < .0001$, was significant, but it was not modulated by ISI, $F[1, 23] = 3.01$, MSE = 2, 794, $\eta_p^2 = 0.11$, $p = .09$. Taken together, the results confirm the presence of repetition effects in baseline. It should be noted though that the current theory assumes that these effects are not reflecting feature binding.

Partial Repetition Costs and Garner Interference

The binding theory predicts that the magnitude of GI should be correlated with the strength of partial repetition costs in filtering, but not with the repetition effect in baseline. To further test this hypothesis, a set of four separate regressions was performed, separately for judgments of chroma and value. Figures 5 and 6 plot the results of these analyses, separately for judgments of chroma and value, respectively. As can be noted, the magnitude of GI was accounted for by the magnitude of partial repetition costs in filtering in both short ($R^2 = 0.50$, $F[1, 20] = 20.64$, $p < .005$) and long ($R^2 = 0.18$, $F[1, 20] = 2.15$, $p < .005$) ISIs. However, no relations, whatsoever, have been documented between these two variables in baseline blocks [$Fs < 1$]. These results confirm a major prediction of the binding theory, namely that to a large extent, GI is manufactured by a binding mechanism in filtering.

Discussion

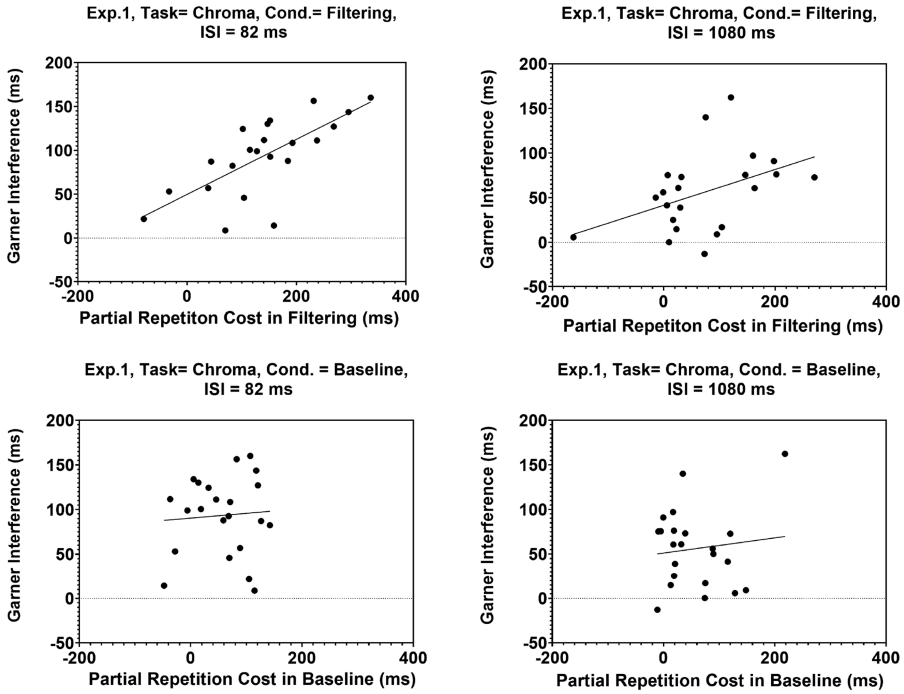
Experiment 1 was set to test the proposed binding theory of GI with the dimensions of chroma and value. Performance with the two dimensions produced comparable results. Substantial levels of GI were observed, in accordance with the expected integral processing. All central hypotheses of the study were supported. First, as predicted, GI decreased as a function of ISI. Second, the conjecture that object-files or even-files binding strength decrease with time was supported. Partial repetition costs (Hommel's, 1998; Hommel, 2004) in filtering were smaller in long ISI compared to short ISI. Third, the magnitude of GI was correlated with the strength of partial repetition costs in filtering but not in baseline. The current theory postulates that partial repetition costs in filtering reflect a genuine binding mechanism of either response–stimulus (event file) or stimulus–stimulus (object file) features (the two cannot be dissociated at this stage). In contrast, partial repetition costs in baseline are merely unidimensional repetition effects, which do not contribute to the emergence of GI. These results should be replicated with another pair of integral dimensions.

Experiment 2

The goal of Experiment 2 was to further generalize the results from Experiment 1 to yet another pair of integral dimensions. In this experiment, I have used the dimensions of width and height of rectangles. These are considered to be classic integral dimensions (Liu et al., 2022; Macmillan & Ornstein, 1998; Niv et al., 2022; Weintraub, 1971).

Figure 5

Experiment 1: Task = Chroma, Garner Interference (ms) as a Function of Partial Repetition Cost (ms) in Filtering (Top Panel) and in Baseline (Bottom Panel) in Short and Long ISIs

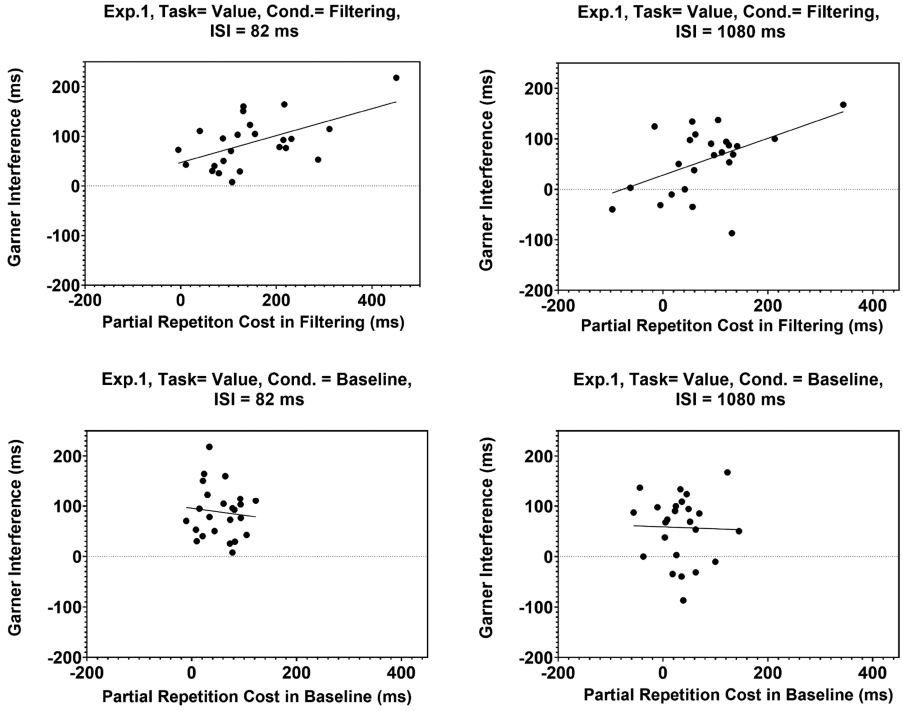


Note. ISIs = inter stimulus intervals.

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Figure 6

Experiment 1: Task = Value, Garner Interference (ms) as a Function of Partial Repetition Cost (ms) in Filtering (Top Panel) and in Baseline (Bottom Panel) in Short and Long ISIs



Note. ISIs = inter stimulus intervals.

Method

Participants

A total of 52 young volunteers from Ariel University participated in return for course credit. Twenty-eight participants ($F = 27$, $M = 1$, $M_{age}{\sim} = 22.03$) were assigned to judge the width of rectangles, whereas the remaining 24 participants ($F = 21$, $M = 3$, $M_{age} = 21.58$) were recruited to classify the height of rectangles. All had normal or corrected-to-normal vision. All gave their signed consent.

Stimuli and Apparatus

The stimuli were four gray rectangles presented over white background. They were created according to a factorial intersection of two levels of width (3 cm or 3.5 cm) and two levels of height (1 cm or 2 cm) (see Figure 7). Viewed from a distance of 56 cm the stimuli subtended approximately 3.3° × 2.04° visual angle.

Procedure and Design

One group of participants performed in classification along the width dimension and ignored the rectangles' height; whereas another group judged the rectangles' height and ignored their width. The experiment consisted of two types of blocks. In Filtering, both width and height varied in an orthogonal fashion. For the group that judged width, the baseline blocks were constructed such that the height was restricted to either 1 cm (Baseline 1) or 2 cm (Baseline 2). For the group that classified the rectangles along the height dimension, the baseline blocks were constructed such that the width was restricted to either 3 cm (Baseline 1) or 3.5 cm (Baseline 2). Before the experiment began, participants engaged in a short training session on the relevant dimension. In this training session, feedback was given to the participants. The remaining details are identical to those reported in previous experiments.

Results

The data of two participants from the width condition were excluded from analysis due to very high error rates (>40%). RTs shorter than 150 ms or longer than 1,800 ms (4.3% and 1.7%, for the width and height conditions, respectively) and error trials (8.9% 8.2% for the width and height conditions, respectively) were removed from analyses.

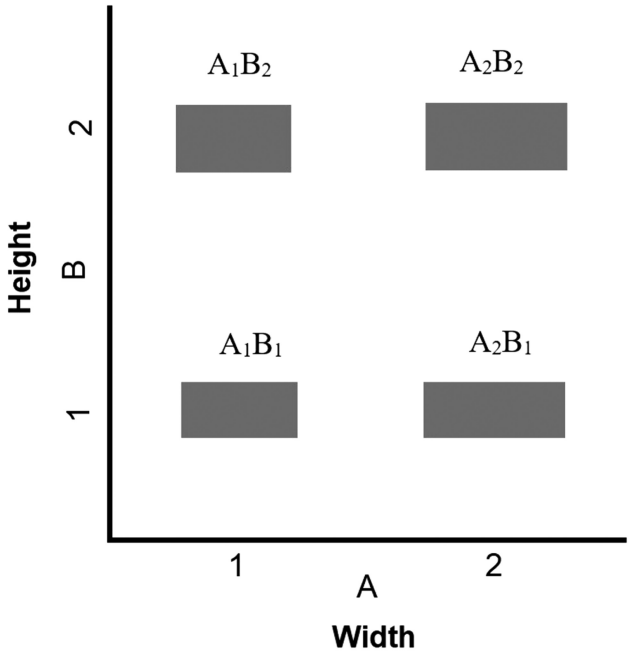
Garner Interference

Across-Tasks

A three-way ANOVA with Task (width, height) × Block (filtering, baseline) and ISI (82 ms, 1,080 ms) was conducted. A main effect of block, $F[1, 144] = 163.4$, MSE = 407, 803, $\eta_p^2 = 0.53$, $p < .0001$, supported the presence of a substantial Garner interference of 90 ms, which was modulated by ISI, $F[1, 144] = 19.94$, MSE = 49, 772, $\eta_p^2 = 0.12$, $p < .0001$. A main effect of ISI, $F[1, 144] = 8.05$, MSE = 20, 106, $\eta_p^2 = 0.05$, $p < .01$, was also significant. No other effects were found to be significant. I

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Figure 7
Experiment 2: Rectangle Stimuli Created by Intersecting Two Levels of Width and Two Levels of Height



conducted two separate ANOVAs on the data of the width and height conditions, to provide a closer look at these effects.

Judgments of Width

The top panel of Figure 8 gives mean RTs and error rates in this task. GI was recorded, $F[1, 25] = 143.6$, MSE = 281, 989, $\eta_p^2 = 0.85$, $p < .0001$, suggesting that the classification of rectangles according to the relevant dimension of width was slower in filtering than in baseline. This outcome reflects the breakdown of selective attention to width in the face of irrelevant variation in height. GI was modulated by ISI, $F[1, 25] = 39.47$, MSE = 38, 396, $\eta_p^2 = 0.61$, $p < .0001$. It was much larger (143 ms) in short ISI, $F[1, 25] = 198.1$, MSE = 264, 247, $\eta_p^2 = 0.88$, $p < .0001$, than in long (66 ms) ISI, $F[1, 25] = 35.04$, MSE = 56, 138, $\eta_p^2 = 0.58$, $p < .0001$. A main effect of ISI was not significant, $F[1, 25] = 4.13$, MSE = 15, 396, $\eta_p^2 = 0.14$, $p = .052$. Comparable analyses on error rates revealed the presence of GI, $F[1, 25] = 61.89$, MSE = 0.163, $\eta_p^2 = 0.71$, $p < .005$.

Judgments of Height

The bottom panel of Figure 8 presents mean RTs and error rates in this task. A sizable GI was documented, $F[1, 23] = 58.3$, MSE = 121, 391, $\eta_p^2 = 0.71$, $p < .0001$, such that classification of rectangles according to the relevant dimension of height was slower in filtering than in baseline. This outcome reflects the breakdown of selective attention to height in the face of irrelevant variation in width. GI was modulated by ISI, $F[1, 23] = 13.91$, MSE = 13, 605, $\eta_p^2 = 0.37$, $p < .005$. It was much larger (95 ms) in short ISI, $F[1, 23] = 61.07$, MSE = 108, 138, $\eta_p^2 = 0.72$, $p < .0001$, than in long (47 ms) ISI, $F[1, 23] = 20.87$, MSE = 26, 859, $\eta_p^2 = 0.47$, $p < .0005$, confirming yet again the predictions of the binding account. The effect of ISI was not significant, $F[1,23] = 1.27$, MSE = 6,099, $\eta_p^2 = 0.05$, $p = 0.27$. Comparable analyses on error rates revealed a GI, $F[1, 23] = 13.42$, MSE = 0.020, $\eta_p^2 = 0.36$, $p < .005$, that was modulated by ISI, $F[1, 23] = 6.51$, MSE = 0.0049, $\eta_p^2 = 0.22$, $p < .05$. Planned comparisons confirmed that GI was larger (4.4%) in short, $F[1, 23] = 15.76$, MSE = 0.022, $\eta_p^2 = 0.40$, $p < .0005$ than in long (1.5%), $F[1, 23] = 3.22$, MSE = 0.0027, $\eta_p^2 = 0.12$, $p = .08$ ISI.

Partial Repetition Costs in Filtering

Across-Tasks

A mixed-design four-way ANOVA with Task (width, height) × Width (repeat, change) × Height (repeat, change) × ISI (short, long) was conducted on mean RTs. Task served as a between-participants factor. A repetition effect of Width, $F[1, 336] = 49.67$, MSE = 235, 111, $\eta_p^2 = 0.13$, $p < .0001$, and a repetition effect of Height, $F[1, 336] = 36.68$, MSE = 173, 619, $\eta_p^2 = 0.10$, $p < .0001$, were significant. Most importantly, indication of partial repetition cost was evident in the form of a significant Width × Height interaction, $F[1, 336] = 63.79$, MSE = 301, 929, $\eta_p^2 = 0.16$, $p < .0001$, which was modulated by ISI, $F[1, 336] = 15.76$, MSE = 74, 631, $\eta_p^2 = 0.04$, $p < .0001$. Other significant interactions included Height × Task, $F[1, 336] = 15.99$, MSE = 75, 697, $\eta_p^2 = 0.105$, $p < .0001$, and Task × Height × ISI, $F[1, 336] = 6.27$, MSE = 29, 691, $\eta_p^2 = 0.02$, $p < .05$. These were not subjected to further analyses due to lack of theoretical interest.

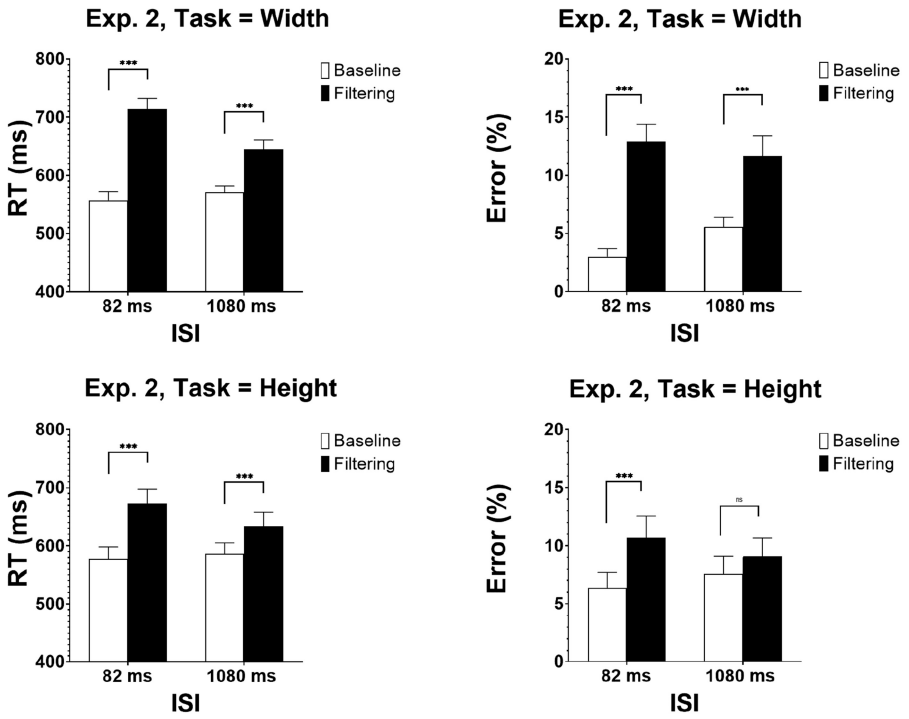
Classifications by Width

The top panel of Figure 9 gives mean RTs and error rates in this task. Repetition effects were observed for both the relevant dimension of width, $F[1, 25] = 9.78$, MSE = 81, 260, $\eta_p^2 = 0.28$, $p < .005$, and irrelevant dimension of height, $F[1, 25] = 58.6$, MSE = 249, 269, $\eta_p^2 = 0.70$, $p < .0001$. Recall that these effects are not consequential for the emergence of binding effects. The effect of ISI was significant too, $F[1, 25] = 19.5$, MSE = 227, 013, $\eta_p^2 = 0.43$, $p < .0001$. Most importantly, evidence for partial repetition costs was provided by a significant Width × Height interaction, $F[1, 25] = 61.34$, MSE = 252, 269, $\eta_p^2 = 0.71$, $p < .0001$, such that complete repetitions and alternations (RR = 643 ms and CC = 647 ms) were responded to faster than partial repetitions (RC = 755 ms, CR = 619 ms), with the exception that complete alteration trials were slower than the partial repetition condition CR. As predicted by the sequential binding account, these partial repetition costs were modulated by ISI, indicated by a significant Width × Height × ISI interaction, $F[1, 25] = 5.60$, MSE = 20, 910, $\eta_p^2 = 0.18$, $p < .05$. This result suggested that binding was stronger in short than long ISIs. In particular, the estimator of binding strength (MIC) for partial repetition costs was larger (MIC = 180 ms) in short than in long (MIC = 98 ms) ISIs, $t[25] = 2.40$, $p < .01$. The Width × ISI, $F[1, 25] = 10.62$, MSE = 20, 546, $\eta_p^2 = 0.29$, $p < .005$, and Height × ISI, $F[1, 25] = 41.26$, MSE = 91, 731, $\eta_p^2 = 0.62$, $p < .0001$, interactions were significant, but were not subjected to further analysis due to lack of theoretical interest. Comparable analyses on error revealed significant Width × Height interaction,

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Figure 8

Experiment 2 (Width and Height of Rectangles): Mean RTs and Error Rates as a Function of Block (Baseline, Filtering) and ISI



Note. Error bars are standard error of the mean. RTs = response times; ISI = inter stimulus interval.
* $p < .05$. ** $p < .01$. *** $p < .0001$.

$F[1, 25] = 5.56$, MSE $= 0.035$, $\eta_p^2 = 0.18$, $p < .05$, which supported the operation of a binding mechanism.

Classification by Height

A repetition effect was observed for the relevant dimension of height, $F[1, 23] = 28.34$, MSE $= 158, 828$, $\eta_p^2 = 0.55$, $p < .0001$, but not for the irrelevant dimension of width, $F[1, 23] = 2.34$, MSE $= 9, 632$, $\eta_p^2 = 0.09$, $p = .14$. The effect of ISI was significant, $F[1, 23] = 7.41$, MSE $= 92, 554$, $\eta_p^2 = 0.24$, $p < .05$. Most importantly, an effect of partial repetition costs was evident in a Width × Height interaction, $F[1, 23] = 72.06$, MSE $= 221, 558$, $\eta_p^2 = 0.75$, $p < .0001$, suggesting that complete repetitions and alternations (RR = 583 ms and CC = 654 ms) were responded to faster than partial repetitions (RC = 665 ms, CR = 708 ms). As predicted by the episodic binding account, the strength of partial repetition costs was modulated by ISI. A significant Width × Height × ISI interaction, $F[1, 23] = 37.6$, MSE $= 57, 558$, $\eta_p^2 = 0.62$, $p < .0001$, supported earlier findings that binding was stronger in short (MIC = 205 ms) than in long (MIC = 66 ms) ISI, $t[23] = 6.13$, $p < .0001$. The Height × ISI interaction was significant, $F[1, 23] = 24.1$, MSE $= 40, 804$, $\eta_p^2 = 0.51$, $p < .0001$, but was not subjected to further analysis due to lack of theoretical interest. Comparable analyses on error revealed an effect of partial repetition costs in the form of a Width × Height interaction, $F[1, 23] = 13.61$, MSE $= 0.010$, $\eta_p^2 = 0.37$, $p < .0001$, that was modulated by ISI, $F[1, 23] = 9.91$, MSE $= 0.022$, $\eta_p^2 = 0.29$, $p < .005$, such that fewer errors were committed in the CC and RR conditions (7.2%, 7.7%) than in the CR and RC conditions (15%, 9.39%). These results replicate those recorded in Experiments 1, and provide strong support for the sequential binding hypotheses regarding the weakening of object-files (event-files) as a function of the time lag ensuing between consecutive trials.

Repetition Effects in Baseline

Repetition effects in Baseline were assessed by a three-way ANOVA with Task (width, height) × Trial (RR, CR) × ISI (short, long) as factors. A main effect of Trial, $F[1, 144] = 21.58$, MSE $= 57, 493$, $\eta_p^2 = 0.13$, $p < .0001$, was the only significant effect. Recall that according to the binding theory of GI, repetition effects in baseline do not reflect a genuine binding process.

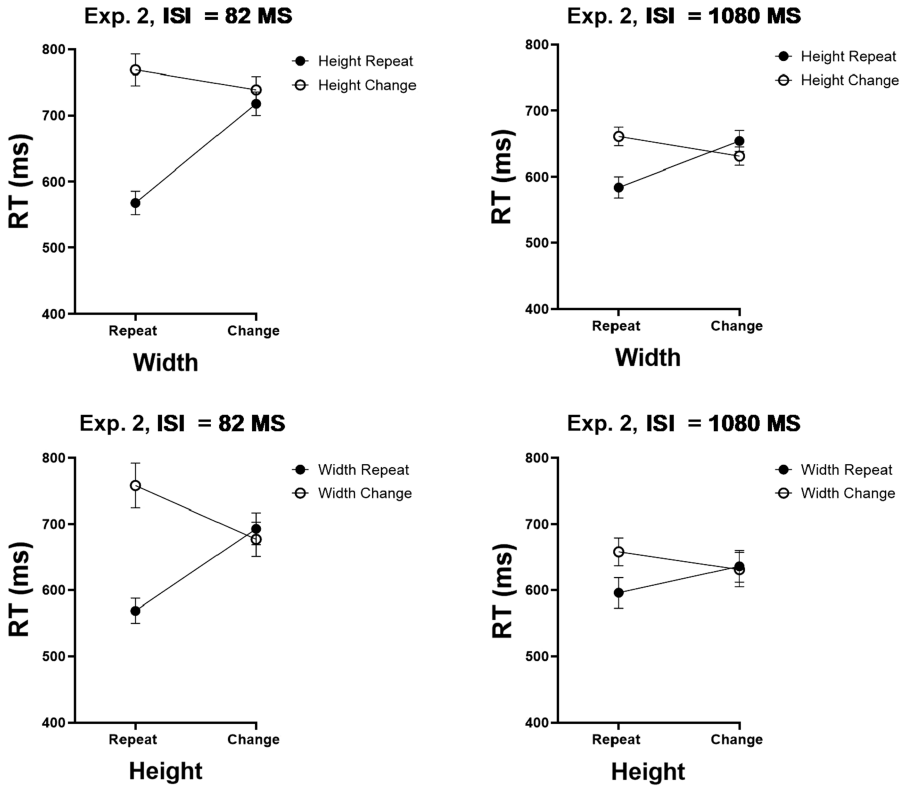
Partial Repetition Costs and Garner Interference

The proposed model predicts that individual differences in GI can be accounted for by the magnitude of partial repetition costs in filtering. Figures 10 and 11 present the results. For judgments of width, there was a tendency for mild correlation between the magnitude of GI and partial repetition cost in short ISI, $R^2 = 0.12$, $F[1, 24] = 3.29$, $p = .08$, but it was only on the verge of significance. There was no sign of correlation in the filtering block with long ISI,

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Figure 9

Experiment 2: Mean RTs in Filtering as a Function of Repetition and Alterations of Features of the Relevant and Irrelevant Dimension at the Two Levels of ISIs



Note. Error bars are standard error of the mean. RTs = response times; ISIs = inter stimulus intervals.

$R^2 = 0.05$, $F[1, 24] = 1.32$, $p = .26$, and in neither of the baseline blocks [$F's < 1$]. One possible explanation for the absence of correlation in judgment of width is a ceiling effect. Low variability in the magnitudes of GI likely masked the expected pattern, as most observers produced relatively comparable large magnitudes of interference. It is plausible that a binding mechanism was active, but a ceiling effect prevented the manifestation of the expected statistical pattern. However, for judgments of height, the magnitude of GI was correlated with the strength of partial repetition costs in filtering, both in short, $R^2 = 0.53$, $F[1, 22] = 25.15$, $p < .0001$) and long ($R^2 = 0.56$, $F[1, 22] = 28.52$, $p < .0001$, ISIs. As expected, this pattern was not observed in the baseline blocks [$Fs < 1$]. These results support the sequential binding hypothesis that bindings processes in filtering plays a central role in generating GI.

Discussion

Experiment 2 was set to test the sequential binding theory of GI with the integral dimensions of rectangles' width and height. The experiment yielded mostly comparable results in the width and height tasks. As predicted, GI diminished in magnitude as a function of ISI. Second, smaller partial repetition costs were documented in long compared to short ISIs in filtering. Third, in judgments of height GI was strongly correlated with the magnitude of partial repetition costs in filtering, strengthening the view that GI is quantitatively related to the amount of stimulus–response or stimulus–stimulus binding in this condition.

Taken together, the results from Experiments 1 and 2 with integral dimensions adduce strong support in the novel binding theory of GI. They provide ample support in the hypothesis that dimensional interaction is driven by feature-binding. However, it is essential to demonstrate that separable and integral dimensions are significantly different with respect to the amount of feature binding they induce. Experiment 3 was designed to test the predictions of the novel binding theory of GI with a pair of separable dimensions (circle size and a diameter's angle drawn within it).

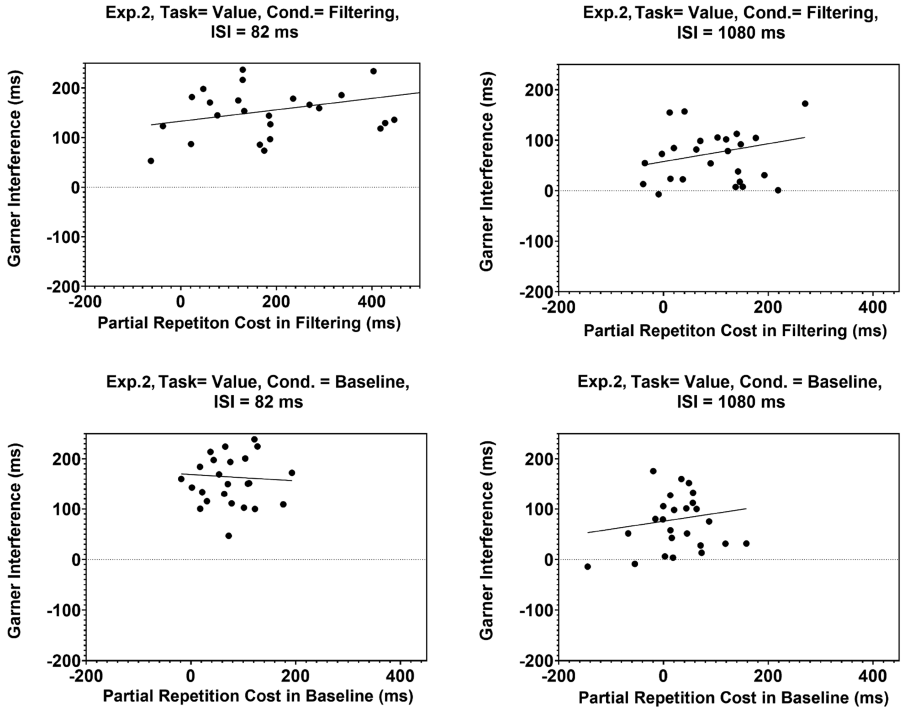
Experiment 3

The dimensions of circle size and angle of diameter were used. These are considered to be classic separable dimensions (Garner & Felfoldy, 1970; Shepard, 1964). In this experiment, one group of participants classified the circles along their size dimension, while ignoring irrelevant variation on the angle of a diameter drawn within them. Another group of participants judged the angle of a diameter while ignoring irrelevant variation on the circle's size. The main prediction entertained is that these separable dimensions will produce minuscule or no GI, and induce significantly smaller partial repetition costs than those exerted by integral dimensions.

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Figure 10

Experiment 2: Judgments of Rectangles Along the Width Dimension



Note. Magnitude of Garner interference (ms) as a function of the binding strength (ms) in filtering (top panel) and in baseline (bottom panel) in short and long ISIs. RTs = response times; ISIs = inter stimulus intervals.

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Method

Participants

A total of 30 young volunteers from Ariel University participated in return for course credit. Of this group, 16 participants ($F = 14$, $M = 2$, $M_{age} = 22.78$) were assigned to judge the angle of diameter, whereas the remaining 14 participants ($F = 13$, $M = 1$, $M_{age} = 23.21$) were assigned to judge the size of circle. All had normal or corrected-to-normal vision. All gave their signed consent.

Stimuli and Apparatus

The stimuli consisted of four circles (see Figure 12) constructed from a factorial intersection of two levels of a diameter's size (0.98 cm, 1.1 cm) and two levels of a diameter's angle (30°, 330°). The angle of the diameter was measured from the vertical, such that it appeared as tilting either clockwise (30°) or counter-clockwise (− 30°). Viewing distance was 56 cm.

Procedure and Design

Procedure and design were similar to those deployed in previous experiments. The circle's size served as an irrelevant dimension for participants who judged the angle of diameter. In this case, the baseline blocks consisted of trials in which the angle varied from trial-to-trial, but the size of the circle was fixed in one value (0.98 cm in Baseline 1 and 1.1 cm in Baseline 2). The angle of diameter served as the irrelevant dimension for participants who classified by the size of the circle. In this case, the baseline blocks consisted of trials in which the circle's size varied, but the angle of tilt was held fixed (30° in Baseline 1 and 330° in Baseline 2). In both types of judgments, the filtering blocks consisted of trials in which both the angle and size varied in an orthogonal fashion from trial-to-trial. The remaining details are identical to those from previous experiments.

Results

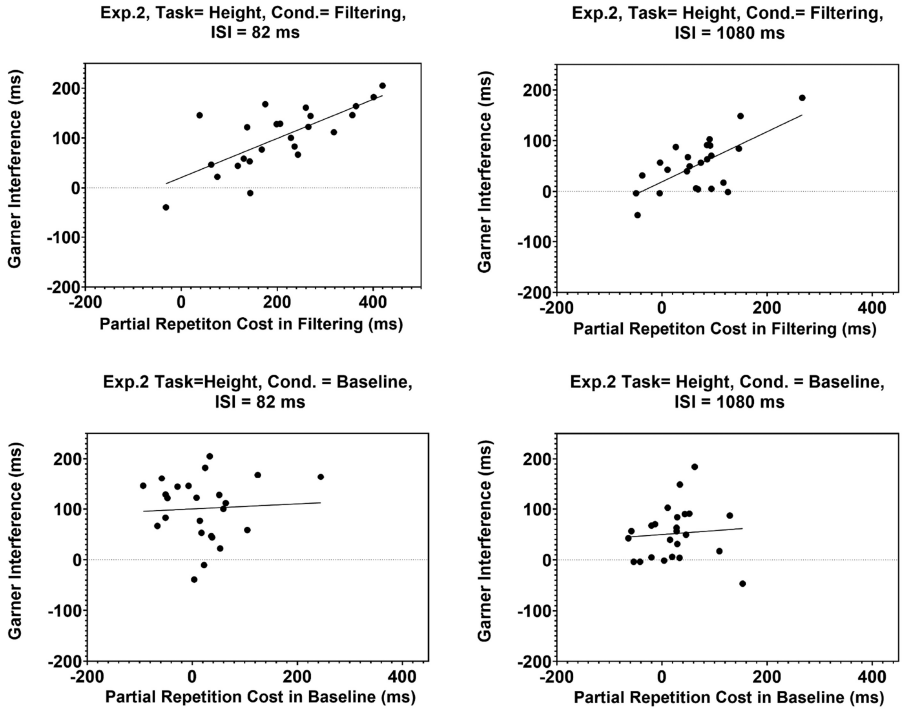
Data can be found here (Fitousi, 2023). RTs shorter than 150 ms or longer than 1,800 ms were removed from analyses. These amounted to 0.56% of the total number of trials. Error trials (3.8% of trials) were also removed.

Garner Interference

A mixed design three-way ANOVA with Task (angle, size) × Block (baseline, filtering) × ISI (short, long) was conducted on mean RTs. Task served as a between-participant factor, whereas Block and ISI served as within-participant factors. The results revealed a main effect of Task, $F[1, 28] = 4.62$, MSE = 2, 834, $\eta_p^2 = 0.14$, $p < .05$, such that classifications of angle (451 ms, $SD\sim = 18.2$) were faster than judgments of size (508 ms, $SD\sim = 19.4$). A main effect of Block, $F[1, 84] = 5.27$, MSE = 3, 231, $\eta_p^2 = 0.06$, $p < .05$, entailed that the performance in filtering was

Figure 11

Experiment 2: Judgments of Rectangles Along the Height Dimension



Note. Magnitude of Garner interference (ms) as a function of degree of binding (ms) in filtering (top panel) and in baseline (bottom panel) in short and long ISIs. ISIs = inter stimulus intervals.

slightly slower than in baseline. This amounted to a minuscule Garner effect of 10 ms. The Task × Block × ISI interaction was not significant, $F[1, 84] = 3.81$, MSE = 2, 337, $\eta_p^2 = 0.04$, $p = .054$. Separate ANOVAs were performed on the data from each type of judgment (angle and size). Figure 13 presents mean RTs and error rates in the baseline and filtering conditions for the two types of judgments (size and angle). These revealed that GI was not present, $F[1, 15] = 3.52$, MSE = 701, $\eta_p^2 = 0.19$, $p = .08$, in judgments of angle, but only in judgments of size, $F[1, 13] = 5.38$, MSE = 2, 816, $\eta_p^2 = 0.29$, $p < .05$.

Moreover, an omnibus ANOVA on the data from Experiments 1–3 with Dimension (integral, separable) × Task (dimension 1, dimension 2) × Block (baseline, filtering) × ISI (short, long), revealed that GI, $F[1, 366] = 86.94$, MSE = 215, 269, $\eta_p^2 = 0.19$, $p < .0001$, was modulated by the type of dimension (separable, integral), $F[1, 366] = 53.03$, MSE = 131, 310, $\eta_p^2 = 0.13$, $p < .0001$, such that GI was eight times larger in judgments of integral (86 ms) than in judgments of separable (10 ms) dimensions, $t[223] = 12.7$, $p < .0001$. This omnibus ANOVA also confirmed the prediction that the decrease in GI with ISI occurs only for integral but not separable dimensions, $F[1, 366] = 8.36$, MSE = 20, 702, $p < .01$. Taken together, these results support the distinction between separable and integral dimensions in the current experiments.

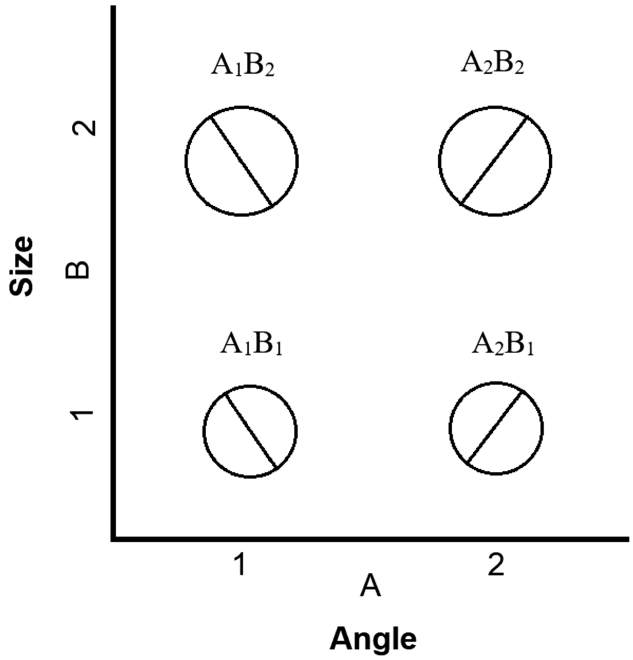
Partial Repetition Costs in Filtering

Binding in the filtering blocks was assessed by a mixed-design four-way ANOVA with Task (angle, size) × Diameter (repeat, change) × Circle (repeat, change) × ISI (short, long) as factors. Task served as a between-participants factor, while the remaining factors served as within-participant factors. A main effect of Task, $F[1, 28] = 6.09$, MSE = 8, 518, $\eta_p^2 = 0.18$, $p < .05$, replicated the finding that angle was judged more efficiently than size. A Circle × Diameter interaction was significant, $F[1, 196] = 13.2$, MSE = 18, 472, $\eta_p^2 = 0.06$, $p < .05$, and modulated by Task, $F[1, 196] = 10.03$, MSE = 14, 020, $\eta_p^2 = 0.05$, $p < .05$. No other effect was significant. These results may suggest the presence of binding costs that are modulated by task. Indeed, a Circle × Diameter interaction was found to be significant for judgments of size, $F[1, 13] = 14.79$, MSE = 30, 319, $\eta_p^2 = 0.53$, $p < .05$, but not for classification of angle [$F < 1$]. The results entail that binding was present only in judgments of size, for which a small but significant GI did emerge. This is consistent with the current theory postulation that GI is dependent on the strength of feature binding. Figure 14 presents the observed patterns as a function of task and ISI. The results entail that binding costs were null in judgments of angle, and present in judgments of the circle's size, amounting to MIC = 99 ms and MIC = 33 ms, in long and short ISI's, respectively. These are relatively reduced magnitudes compared to those observed with integral dimensions.

An omnibus ANOVA with Dimensions Type (integral, separable) × Dimension 1 (repeat, change) × Dimension 2 (repeat, change) × ISI (short, long) supported the conjecture that strength of binding, $F[1, 868] = 69.43$, MSE = 299, 362, $p < .0001$, was modulated by the type of dimensions (integral, separable) involved, $F[1, 868] = 21.99$, MSE = 948, 409, $p < .0001$. Planned comparisons showed that mean strength of binding with integral dimensions

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Figure 12
The Stimuli Set Used in Experiment 3



Note. These were created by a factorial intersection of two levels of diameter's angle and two levels of circle size.

was almost four times larger (126 ms) than binding strength with separable dimensions (35 ms) ($t[175] = 7.25$, $p < .00001$). Moreover, binding was modulated by ISI, as suggested by a Dimension 1 × Dimension 2 × ISI significant interaction, $F[1, 868] = 21.99$, MSE = 94, 840, $p < .0001$, supporting the conjecture that feature binding strength decreases with time.

Repetition Effects in Baseline

Repetition effects in baseline were assessed using a mixed-design three-way ANOVA with Task (angle, size) × Trial (RR, CR) × ISI (short, long). A main effect of Trial, $F[1, 84] = 4.1$, MSE = 5, 315, $\eta_p^2 = 0.05$, $p < .05$, was the only significant effect. To assess the relation between separable and integral dimensions with respect to the strength of repetition effects in baseline, an omnibus ANOVA with Dimension type (integral, separable) × Trial (repeat, change) × ISI (short, long) was conducted. This ANOVA pointed to a main effect of Trial, $F[1, 372] = 6.77$, MSE = 188, 525, $p < .01$, which was modulated by the type of dimensions (integral, separable), $F[1, 372] = 27.50$, MSE = 76, 509, $p < .0001$. Planned comparisons showed that the repetition effect in baseline was stronger for integral (43 ms) than separable (14 ms) dimensions ($t[123] = 7.83$, $p < .01$). In addition, across all experiments, the effect of trial (partial repetition cost) in baseline was not modulated by ISI [$F < 1$]. Recount that repetition effects in baseline do not reflect a binding mechanism.

Partial Repetition Costs and Garner Interference

The novel binding account proposed here predicts that in the case of separable dimensions, where GI is expected to be minuscule or nonexistent, the strength of feature-binding should not be correlated with the magnitude of GI neither in filtering nor in baseline. To further test this hypothesis, a set of eight separate regressions were performed for judgments of angle and size. Figures 15 and 16 plot the results of these analyses. First, in judgments of angle, the magnitude of GI was not accounted for by the strength of binding effect in filtering in both short, $R^2 = 0.21$, $F[1, 15] = 3.73$, $p = .07$, and long $R^2 = 0.10$, $F[1, 15] = 1.69$, $p = .21$, ISIs. Similarly, in judgments of size, the magnitude of GI was not accounted for by the magnitude of binding in filtering, neither in short, $R^2 = 0.067$, $F[1, 12] = 3.73$, $p = .07$, or long, $R^2 = 0.23$, $F[1, 12] = 3.66$, $p = .07$, ISIs. A comparable set of analyses were conducted to assess the relations between GI and repetition effects in the baseline blocks. None of these analyses yielded a significant result (all $Fs < 1$).

General Discussion

Summary and Contribution

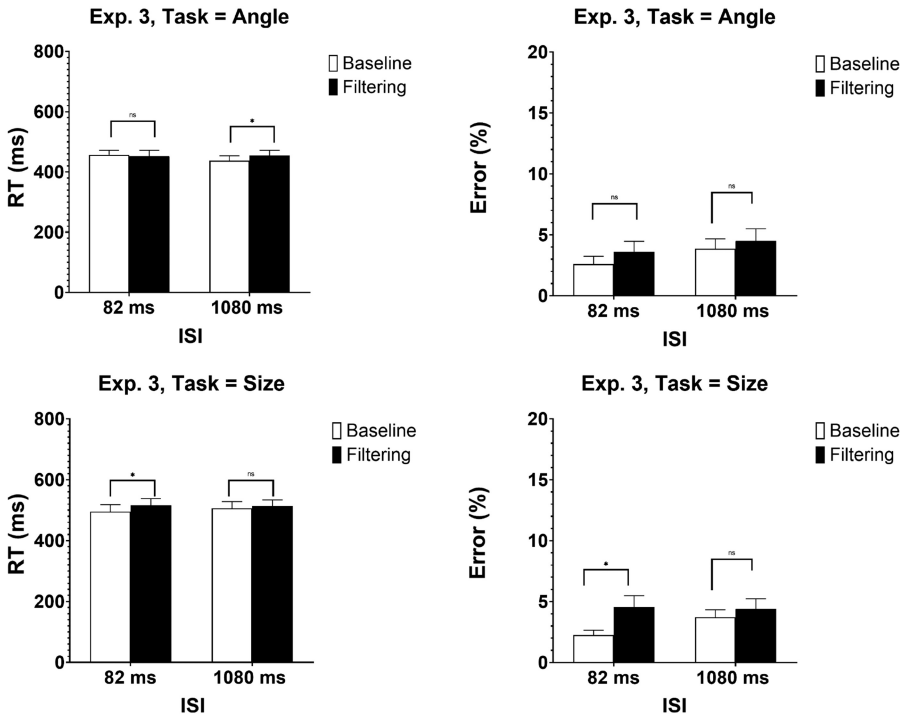
The distinction between integral and separable dimensions is fundamental for many areas of cognitive research (for review, Algom & Fitousi, 2016), and GI is considered to be one of Psychology's powerful tools for assessing the separability–integrality distinction, as well as their associated patterns of selective attention (Algom & Fitousi, 2016). A great deal of research has been dedicated to uncovering the mechanisms that generate GI (Ashby & Maddox, 1994; Burns, 2016; Dyson & Quinlan, 2010; Little et al., 2016; Niv et al., 2022; Nosofsky & Palmeri, 1997), but no consensus has been reached so far with respect to the nature of its origins (for recent discussions see, Burns, 2016; Niv et al., 2022). The current study proposes a novel theory of GI, one that builds on earlier well-established notions of feature integration and object files (Hommel's, 1998; Kahneman et al., 1992). The theory is unique because it ascribes the interference solely to feature binding processes at the micro (trial-to-trial) level. According to this account, GI is generated through a series of binding processes and feature competition across sequential trials. The account assumes that participants integrate relevant and irrelevant features into an episodic object file or an event file on a trial-by-trial basis. GI is the by-product of partial repetition costs in trials that present competition among features from previously stored conjunctions. Formal analytic decomposition revealed that the magnitude and very existence of GI depends on these feature binding and competition processes in filtering. The theory received substantial empirical support in three experiments. Experiments 1 and 2 used pairs of integral dimensions (chroma and value, height and width of rectangles), whereas Experiment 3 harnessed a pair of separable dimensions (circle's size and diameter's angle). The results of all three experiments converged, adducing ample support in the binding account.

The current study underscored various significant differences in the extent to which feature binding manifested with integral and separable dimensions. First, as expected, integral dimensions produced substantial amounts of GI that decreased as a function of ISI; whereas separable dimensions produced minuscule or no GI. Second, sizable amounts of partial repetition costs—accepted signatures of feature binding—were documented with integral dimensions in filtering. In contrast, with separable dimensions, comparatively reduced or no such partial repetition costs were

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Figure 13

Experiment 3 (Diameter's Tilt and Circle's Size): Mean RTs and Error Rates as a Function of Block (Baseline, Filtering) and ISI



Note. Error bars are standard error of the mean. RTs = response times; ISI = inter stimulus interval. * $p < .05$. ** $p < .01$. *** $p < .0001$.

documented. Moreover, the strength of GI with integral dimensions was correlated with the magnitude of partial repetition costs in filtering, but not in baseline. No such effects were documented with separable dimensions. Moreover, for integral dimensions, partial repetition costs diminished as a function of ISI, whereas for separable dimensions they did not. Taken together, these results strongly support the view of the binding theory, namely that integral and separable dimensions produce different patterns of GI due to the operation of stronger feature bindings and feature competition. The concurrence of these phenomena is not incidental, but rather reflects a joint time-dependent memory mechanism. The upshot is that GI (=dimensional interaction) is driven by feature binding.

Sequential Binding and Redundancy Gains

The current study focused exclusively on explaining GI. But the sequential binding model can account for other related Garner phenomena. For example, the presence of *redundancy gains* in *Correlation* (Fitousi, 2021; Garner, 1974; Melara & Mounts, 1993; Pansky & Algom, 2002; Pomerantz, 1983). Recall that in correlation, values of the relevant dimension vary in tandem with values of the irrelevant dimension. With integral dimensions, performance in correlation is often improved relative to baseline (i.e., redundancy gain) (Garner & Felfoldy, 1970), a pattern that, together with GI, provides a set of converging operations on the notion of integrality (Fitousi, 2013; Garner et al., 1956). The sequential feature binding account can readily explain this facilitation by adhering to: (a) basic principles of feature integration (Hommel's, 1998) and (b) the unique structure of sequential trials in correlation. Note that in the Garner design there are two possible correlation blocks. One includes stimuli A_1B_1 and A_2B_2. The second includes stimuli A_1B_2 and A_2B_1. Because the features from the relevant and irrelevant dimensions are correlated, none of the resulting sequences consists of partial repetition trials (RC or CR). In a matter of fact, the correlated blocks contain only complete repetitions RR (e.g., $A_1B_1 \rightarrow A_1B_1$) or complete alterations CC (e.g., $A_1B_1 \rightarrow A_2B_2$) trials. Thus, in correlation, participants do not pay partial repetition costs. Consequently, binding and retrieval from episodic memory are held more efficiently than in baseline.

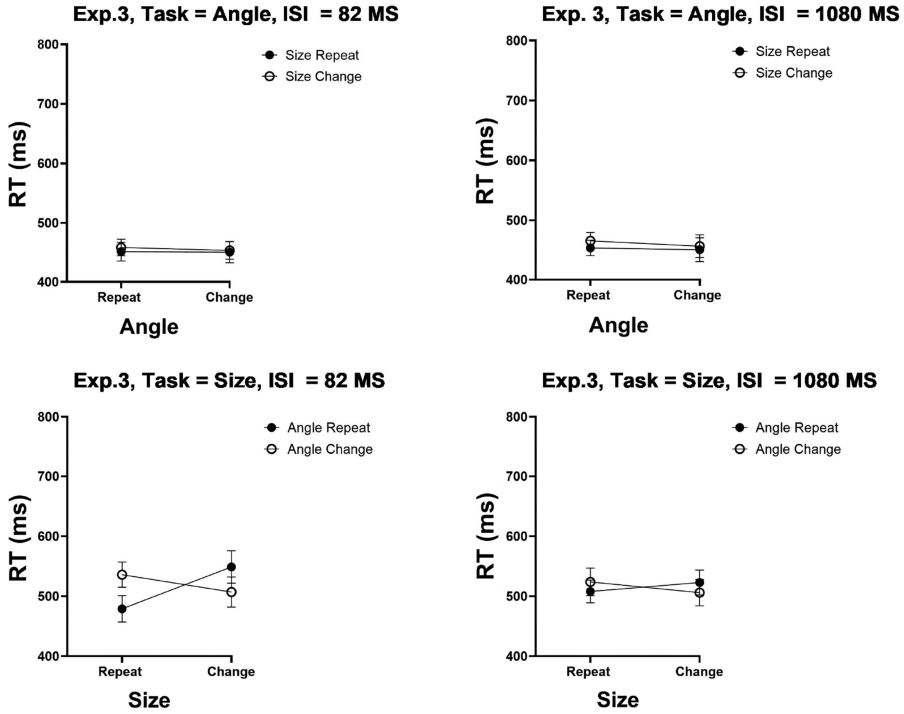
Sequential Binding and the Bypass Strategy

Another point that deserves a comment concerns the nature of the binding structures documented in the current study. Hommel's (1998) has distinguished between "object files" and "event files" (see also, Fitousi, 2017b, 2018). The former are created through binding of stimulus features (e.g., color and shape), whereas the latter are generated via integration of response and stimulus features (response and shape) (Fitousi, 2017a, 2017b; Frings et al., 2013; Frings & Rothermund, 2011; Kühn et al., 2011). Both types of representations are indicated by partial repetition costs. With "event files," for example, participants are faster to respond to the color dimension when both response and color are repeated (left-key + red → left-key + red) or both alternated (left-key + red → right-key

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Figure 14

Experiment 3: Judgments of Circle's Size and Diameter's Angle. Mean RTs in Filtering as a Function of Repetition and Alterations of Features of the Relevant and Irrelevant Dimension at the Two Levels of ISI



Note. Error bars are standard error of the mean. RTs = response times; ISI = inter stimulus interval.

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+ green) compared to a condition in which one repeats and the other alternates (left-key + red → right-key + red). In the Garner paradigm, the level of the relevant dimension is confounded with its corresponding response. A change in the relevant feature also implies a change in the required response. Thus, the binding effects observed here may be interpreted as response–stimulus (event files) rather than stimulus–stimulus (object files) binding. Previous studies adduced evidence for the impact of ISI on the formation of event files (Hommel & Frings, 2020), but not object files (Hommel, 2005; Hommel & Colzato, 2004). Given that ISI did show a strong impact on the strength of bindings in the current study, it is theoretically plausible that the results reflect the formation of event files rather than object files. However, this conclusion should be delayed until more decisive experiments are conducted.

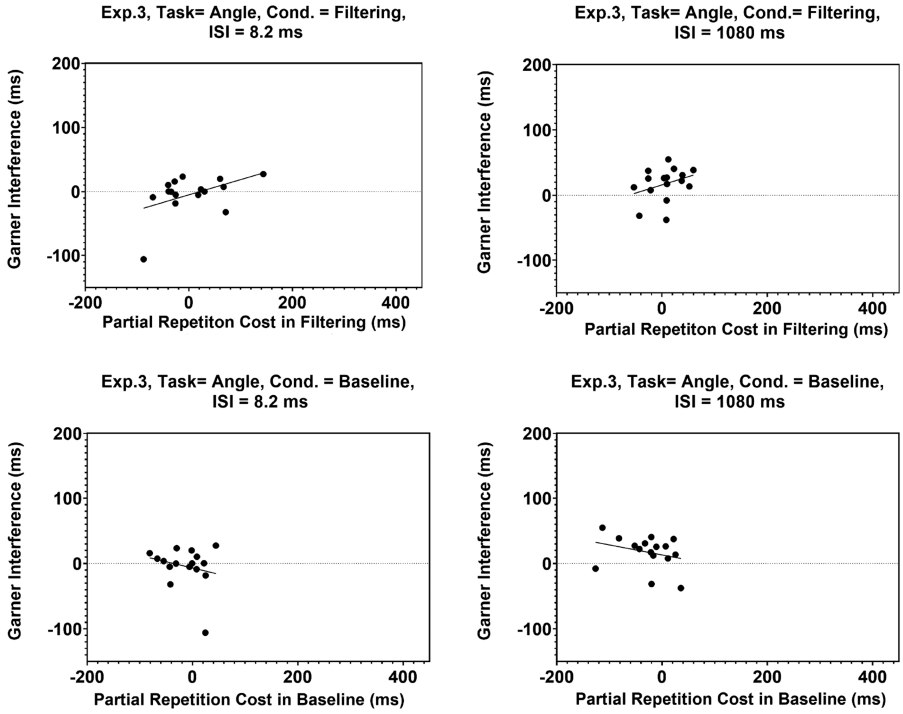
The distinction between "object files" and "event files" can assist us in relating our current notions and findings to previous ideas about sequential effects. The various types of sequential effects that have been proposed in the past to explain GI can now receive a new interpretation. Garner (1974) suggested three species of such effects: (a) a quick-check strategy, according to which participants change their response only if they notice that the stimulus has changed, and (b) an interference that is created when stimuli change on successive trials even when the response remains constant (i.e., S changes & R repeats), a type of sequence that is unique to filtering (c) repetition effects in which the same stimulus repeats, with more repetitions in baseline than in filtering. All three types of effects can be given a novel interpretation in terms of episodic bindings of either stimulus–stimulus integration (i.e., object files) or stimulus–response integration (i.e., event files). First, the quick-check ("bypass") strategy likely relies on the ease with which an object file that is encoded on trial $n-1$ is being retrieved on trial n. In case of partial (features) repetition, the participant experiences difficulty due to feature competition in retrieving the object file, and thus changes her response. If a complete match occurs, no feature competition ensues, retrieval is easy and the previous response is also retrieved. Second, the "S changes & R repeats" effect can be interpreted as resulting from the formation of event files. The participant encodes the stimulus and response as an event file, and if the next trial consists of partial repetition then a cost is incurred on performance. Third, repetition effects can be explained away as benefits due to complete feature repetition (object files).

Sequential Binding and Range Effects

It has been repeatedly shown that expanding the range of values along the irrelevant dimension results in larger GI (Burns, 2016; Huettel & Lockhead, 1999; Melara & Mounts, 1993). What is the mechanismresponsible for this phenomenon? Huettel and Lockhead (1999) conjectured that "…performance is poor when the irrelevant range is large and because successive differences between stimuli within a task are large, on average, when the range is large" (p. 1624). But how exactly successive differences between stimuli lead to larger interference? The current sequential binding account offers a straightforward explanation. Note that a

Figure 15

Experiment 3: Judgment of Angle



Note. Magnitude of Garner interference (ms) as a function of degree of binding (ms) in filtering (top panel) and in baseline (bottom panel) in short and long ISIs. ISIs = inter stimulus intervals.

by-product of adding more irrelevant features to filtering is: (a) an increase in the number of partial (feature) repetition trials, and (b) a decrease in the number of complete (feature) repetition trials. Both (a) and (b) are associated with higher bindings costs, which according to the current formal and empirical investigations, result in larger GI. It is important to note that the sequential binding account assumes that sequential feature integration occurs with separable dimensions as well. However, the strength of binding with those dimensions is substantially weaker.

Sequential Binding and Niv et al.'s (2022) "Roving Baseline"

Recently, Niv et al. (2022) have tested the conjecture that GI results from greater information load in filtering. According to their hypothesis, in the baseline block, no information is transmitted because the irrelevant dimension is held fixed. In contrast, in the filtering block, a great deal of information is transmitted, because the irrelevant dimension varies from trial-to-trial and is therefore uncertain. This difference in information load may account for GI. To test this conjecture, Niv et al. (2022) have added a "roving baseline" condition, in which the value of the irrelevant dimension on trial n was completely determined by its value on trial $n-1$, as is the case in regular baseline, but the values varied from trial-to-trial (as in filtering). A sizeable GI was documented in this roving baseline, which supported the hypothesis that irrelevant variation and not informational load is responsible for GI. The novel sequential binding account presented here can readily explain this finding. Due to its unique nature, the "roving baseline"consists of a larger number of *partial feature repetition* trials than in a regular filtering, and a smaller number of *complete feature repetition* trials (i.e., irrelevant feature on trial $n-1$ must change on trial n). This entails that the "roving baseline" is actually a more challenging block, as it requires more efforts from the participants at retrieval and creation of object files. Thus, it is expected that performance in the "roving baseline" should be similar or even worse than in traditional filtering, as indeed was found by Niv et al. (2022).

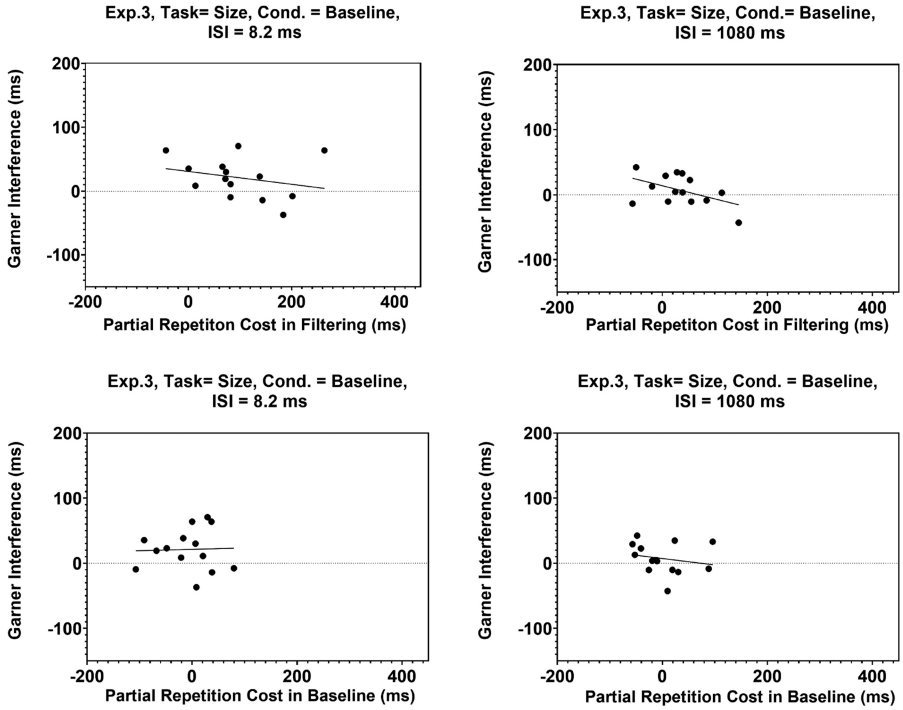
Dimensions Versus Features

Dimensions figure prominently in studies of the Garnerian tradition, while features are often considered to be the elemental building blocks of perception in theories emanating from Treisman and Gelade (1980) *Feature Integration Theory* (see e.g., Hommel's, 1998; Hommel, 2004). The relations between dimensions and features bears tremendous import for theories of attention and categorization (Makovski & Tsal, 2002). However, the distinction between features and dimensions is not always clear. A well-accepted conception is that features are levels that constitute a dimension (Garner, 1976). For example, red and green are possible features of the color dimension. In the visual search literature, it has been shown that observers can attend selectively to features and to dimensions. For example, Müller et al. (1995) found that searching for three targets within a single dimension (e.g., orientation) is faster than searching for three targets defined across three different dimensions. This within-dimension search benefit has been explained in terms of a *dimension-weighting* account. Same-dimensions signals receive

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Figure 16

Experiment 3: Judgment of Circle' Size



Note. Magnitude of Garner interference (ms) as a function of degree of binding (ms) in filtering (top panel) and in baseline (bottom panel) in short and long ISIs. ISIs = inter stimulus intervals.

higher weight on a master map of summed dimensions-specific salience signals. In GRT (Ashby & Maddox, 1994; Ashby & Townsend, 1986), dimensional independence is dissociated from featural independence. The former relates to differences in performance across-objects (defined as conjunctions of features), while the latter is ascribed to interactions within a specific object (Fitousi, 2018, 2021). The distinction between features and dimensions is also important for theories of similarity. Shepard (1962, 1964) proposed a dimensional metric of similarity based on distances between objects on a multidimensional psychological space. According to this model, similarity scaling with integral dimensions is held according to their Euclidean distance in that metric space, whereas the similarity of separable dimensions is computed according to a city-block metric (Garner, 1974). However, Tversky (1977) argued that dimensional measures of similarity violate critical theorems of distance, and proposed instead a feature model. According to this model similarity of two objects is computed as a difference between their common and disjoint features. The upshot is that the distinction between features and dimensions continues to be an important topic in many areas of cognition, and the current study offers a possible theoretical bridge between these two types of representations.

Theoretical Implications for the Integrality/Separability Distinction

The sequential binding account offers several novel contributions: (a) it is a theory of GI and related Garner phenomena. In this framework, all types of sequential effects in the Garner task are interpreted as manifestations of a common processing mechanism of feature binding, (b) several researchers questioned the possibility that sequential effects can explain the integrality/separability distinction because the same trial sequences exist in both (Algom & Fitousi, 2016; Burns, 2016). The current approach assumes that sequential binding occurs with both integral and separable dimensions (Fitousi, 2018; Hommel's, 1998), and that binding strength is what makes them produce different patterns of performance. Thus, integral and separable dimensions differ only quantitatively not qualitatively, (c) the account provides the often missing link between selective attention and memory (Logan et al., 2021). Attention is derived by retrieval of information from memory. When selective attention fails, it is primarily because memory processes were hampered, (d) the account places the integral/separable distinction within the context of binding and feature integration theories, expanding our understanding of how features are integrated in the Garner paradigm, and how dimensions interact, (e) the marriage of selective attention and binding allows us to import related notions from each fields of study. For example, Hommel's work indicates that binding can occur even when the dimensions do not inhere the same location (Fitousi, 2018; Hommel's, 1998). Binding mechanisms within the Garner paradigm can explain situations in which GI is observed with pairs of dimensions that do not inhere the same spatial location, or when cross-modality dimensions are tested (Melara, 1989). Moreover, the degree of binding in the Garner paradigm and the extent to which selectivity is compromised can be explained by the intentional weighting (Memelink & Hommel,

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2013) principle. According to this idea, intention- or goal-related feature dimensions are weighted more strongly, so that feature values along the irrelevant dimension have stronger impact on information binding, and therefore, on the quality of selective attention.

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(*Appendices follow*)

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Appendix A

Table A1 presents all possible types of sequential trials in filtering and baseline blocks in the Garner design. As can be noted the features on both dimensions can either repeat (*R*) or change (*C*), giving rise to four types of trials—RR, CC, RC, and CR (see also Dyson & Quinlan, 2010).

Table A1
Types of Sequential Trials in Filtering and Baseline Conditions

Preceding stimulus	Present stimulus	Dim A	Dim B	Complete match/ mismatch
Filtering				
A_1B_1	A_1B_1	R	R	+
A_1B_1	A_1B_2	R	C	−
A_1B_1	A_2B_1	C	R	−
A_1B_1	A_2B_2	C	C	+
A_1B_2	A_1B_1	R	C	−
A_1B_2	A_1B_2	R	R	+
A_1B_2	A_2B_1	C	C	+
A_1B_2	A_2B_2	C	R	−
A_2B_1	A_1B_1	C	R	−
A_2B_1	A_1B_2	C	C	+
A_2B_1	A_2B_1	R	R	+
A_2B_1	A_2B_2	R	C	−
A_2B_2	A_1B_1	C	C	+
A_2B_2	A_1B_2	C	R	−
A_2B_2	A_2B_1	R	C	−
A_2B_2	A_2B_2	R	R	+
Baseline 1				
A_1B_1	A_1B_1	R	R	+
A_1B_1	A_2B_1	C	R	−
A_2B_1	A_1B_1	C	R	−
A_2B_1	A_2B_1	R	R	+
Baseline 2				
A_1B_2	A_1B_2	R	R	+
A_1B_2	A_2B_2	C	R	−
A_2B_2	A_1B_2	C	R	−
A_2B_2	A_2B_2	R	R	+

Note. Dim A = dimension A (relevant), Dim B = dimension B (irrelevant). R = feature repeat, C = feature change. + = both features repeat or both change, − = one feature repeats while the other changes.

(*Appendices continue*)

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Appendix B

Proof of Equation 6

We can express the degree of Garner interference in terms of binding in filtering by using the decomposition equation offered by Dyson and Quinlan (Equation 3). This equation can be rearranged to express the size of the Garner interference. I also simplified the notation:

$$\mathrm{GI} = \frac{E(\mathrm{CR}_f)}{4} + \frac{E(\mathrm{RC}_f)}{4} + \frac{E(\mathrm{RR}_f)}{4} + \frac{E(\mathrm{CC}_f)}{4} - \frac{E(\mathrm{CR}_b)}{2} - \frac{E(\mathrm{RR}_b)}{2}, \tag{B1}$$

where *R* and *C* stand for repeat or change of the relevant or irrelevant dimension, and the subscripts *f* and *b* stand for filtering and baseline, respectively. This can be rewritten as:

$$\begin{aligned} 4 \times \mathrm{GI} = {} & E(\mathrm{CR}_f) + E(\mathrm{RC}_f) + E(\mathrm{RR}_f) + E(\mathrm{CC}_f) \\ & - 2 \times E(\mathrm{CR}_b) - 2 \times E(\mathrm{RR}_b). \end{aligned} \tag{B2}$$

Now we can subtract the quantity $-2 \times E(\mathrm{RR}_f) - 2 \times E(\mathrm{CC}_f)$ from both left- and right-hand sides of the equation:

$$\begin{aligned} & 4 \times \mathrm{GI} - 2 \times E(\mathrm{RR}_f) - 2 \times E(\mathrm{CC}_f) = E(\mathrm{CR}_f) + E(\mathrm{RC}_f) + E(RR_f) \\ & + E(\mathrm{CC}_f) - 2 \times E(\mathrm{CR}_b) - 2 \times E(\mathrm{RR}_b) - 2 \times E(\mathrm{RR}_f) - 2 \times E(\mathrm{CC}_f). \end{aligned} \tag{B3}$$

Rearranging we get:

$$\begin{aligned} & 4 \times \mathrm{GI} - 2 \times E(\mathrm{RR}_f) - 2 \times E(\mathrm{CC}_f) = E(\mathrm{CR}_f) \\ & + E(\mathrm{RC}_f) - E(\mathrm{RR}_f) - E(\mathrm{CC}_f) - 2 \times [E(\mathrm{CR}_b) + E(\mathrm{RR}_b)]. \end{aligned} \tag{B4}$$

Now, if we focus on the right-hand side of the equation we can see that the first four elements are equal to the amount of partial repetition cost in filtering (Equation 4), whereas the last two elements equal twice the performance in baseline (Equation 3). Therefore, we can rewrite the equation as:

$$4 \times \mathrm{GI} - 2 \times E(\mathrm{RR}_f) - 2 \times E(\mathrm{CC}_f) = E(\mathrm{cost}|\mathrm{filtering}) - 2 \times E(\mathrm{Baseline}) \tag{B5}$$

with *cost* denoting the partial repetition costs. From this, we get that:

$$4 \times \mathrm{GI} = E(\mathrm{cost}|\mathrm{filtering}) - 2 \times E(\mathrm{Baseline}) + 2 \times [E(\mathrm{RR}_f) + E(\mathrm{CC}_f)]. \tag{B6}$$

Now we can write the Garner interference as:

$$\begin{aligned} \mathrm{GI} = {} & E(\mathrm{RT}|\mathrm{cost};\ \mathrm{filtering})/4 - E(\mathrm{RT}|\mathrm{Baseline})/2 \\ & + [E(\mathrm{RT}|\mathrm{RR};\ \mathrm{filtering}) + E(\mathrm{RT}|\mathrm{CC};\ \mathrm{filtering})]/2. \end{aligned} \tag{B7}$$

Received March 22, 2022
Revision received January 10, 2023
Accepted March 15, 2023 ■

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